PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 6 Submission

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Section	G
Department	CSE
Campus	RR

Topic: Lab 6: BrowserForensics

UE23CS342BA6

Objective:

The objective of this experiment is to understand the fundamentals of browser forensics using forensic tools. In this activity you will examine the cache maintained by Google Chrome using NirSoft's ChromeCacheView and ChromeHistoryView.

Tools: Product: ChromeCacheView

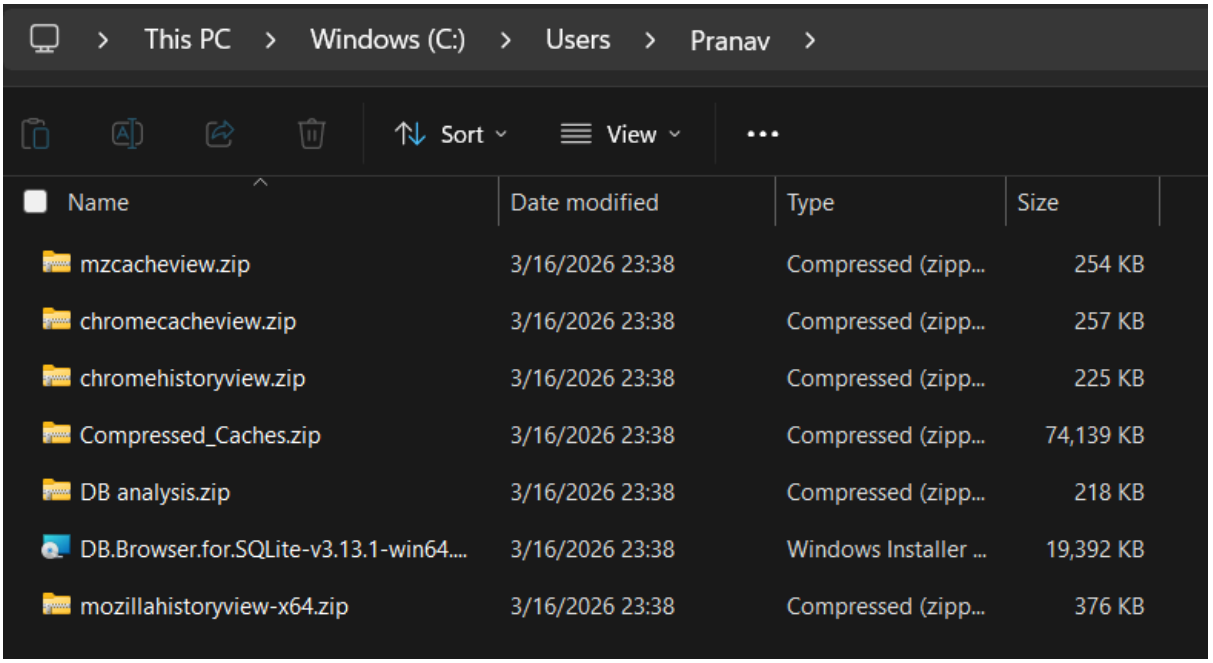
Manufacturer: NirSoft
Website: http://www.nirsoft.net/web browser tools.html

Product: ChromeHistoryView
Manufacturer: NirSoft
Website: http://www.nirsoft.net/web browser tools.html

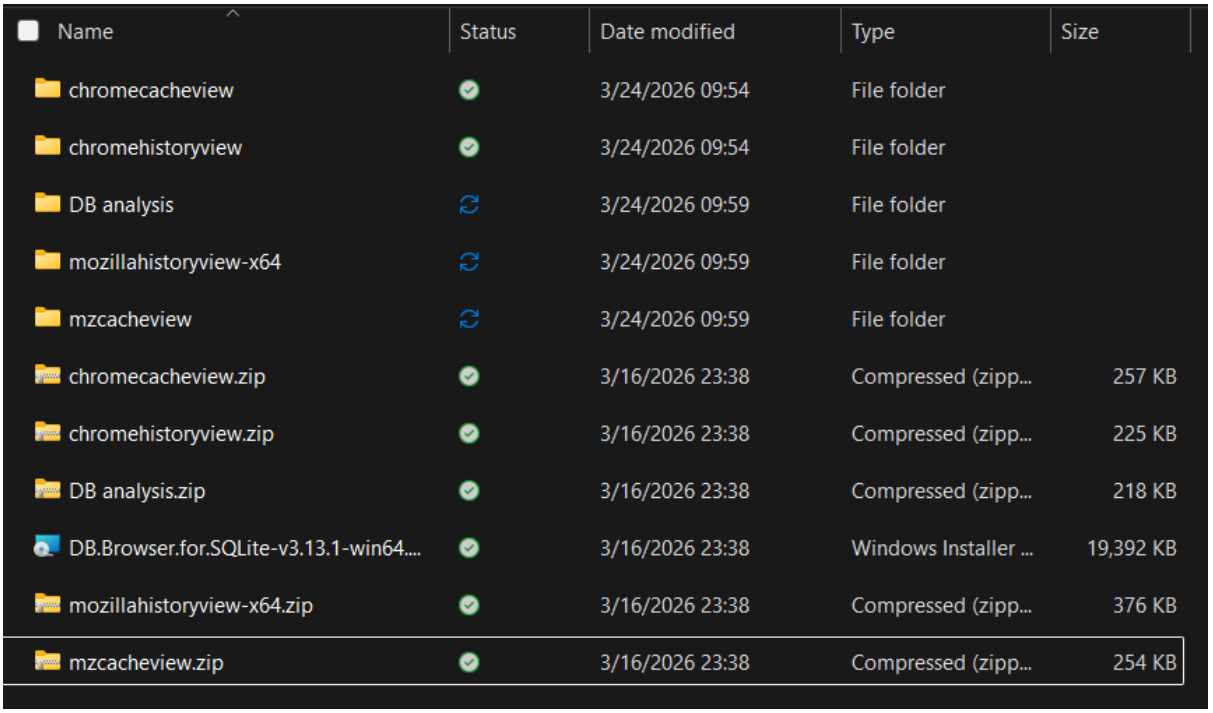
Activity 1: Using ChromeCacheView

Step1: Do the following for ChromeCacheView

1) Download ChromeCacheView. Extract the executable from the compressed file and place it on the desktop.

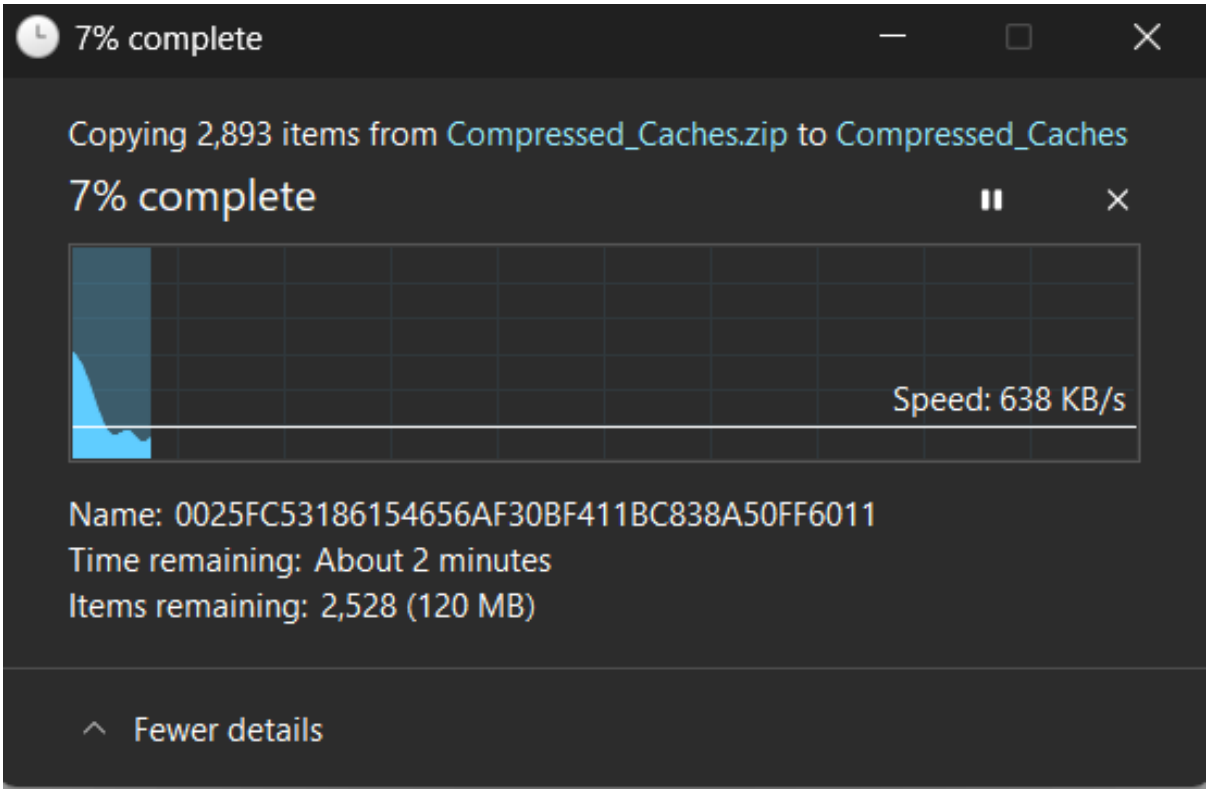


2) Download ChromeHistoryView. Extract the executable from the compressed file and place it on the desktop.



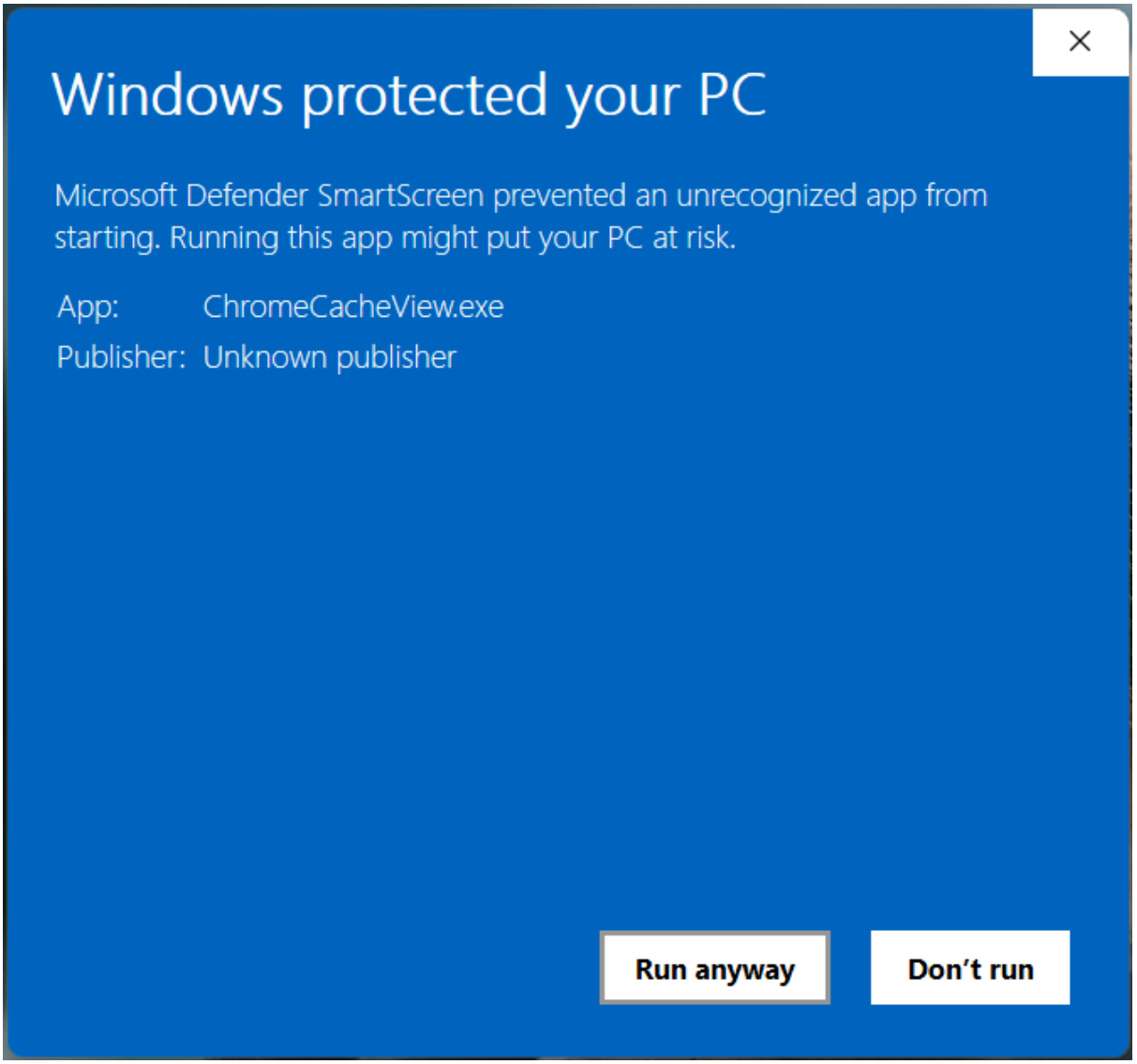
Name	Status	Date modified	Type	Size
chromecacheview	✓	3/24/2026 09:54	File folder	
chromehistoryview	✓	3/24/2026 09:54	File folder	
DB analysis	↻	3/24/2026 09:59	File folder	
mozillahistoryview-x64	↻	3/24/2026 09:59	File folder	
mzcacheview	↻	3/24/2026 09:59	File folder	
chromecacheview.zip	✓	3/16/2026 23:38	Compressed (zipp...	257 KB
chromehistoryview.zip	✓	3/16/2026 23:38	Compressed (zipp...	225 KB
DB analysis.zip	✓	3/16/2026 23:38	Compressed (zipp...	218 KB
DB.Browser.for.SQLite-v3.13.1-win64....	✓	3/16/2026 23:38	Windows Installer ...	19,392 KB
mozillahistoryview-x64.zip	✓	3/16/2026 23:38	Compressed (zipp...	376 KB
mzcacheview.zip	✓	3/16/2026 23:38	Compressed (zipp...	254 KB

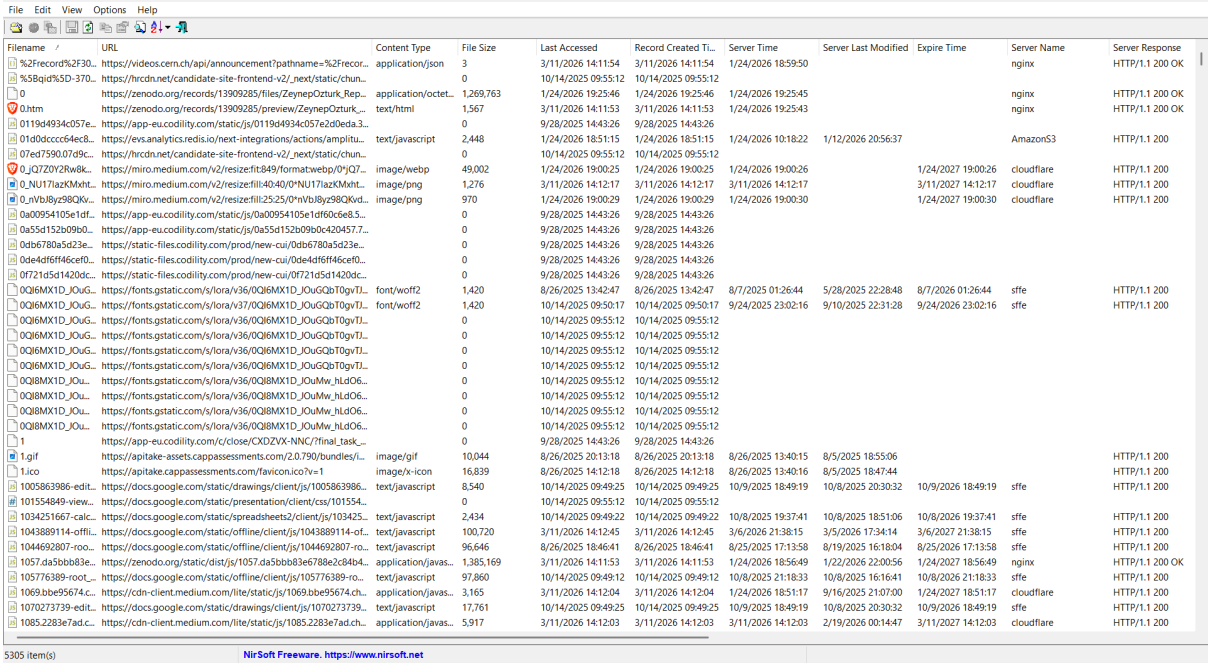
3) Download the file named "Compressed Caches.zip" given and extract the folders to your desktop. (Keep the folder structures intact.)



4) The cache for Google Chrome was extracted from the following location: C:\Users\admin\AppData\Local\Google\Chrome\User Data\default\Cache\

5) Launch ChromeCacheView.





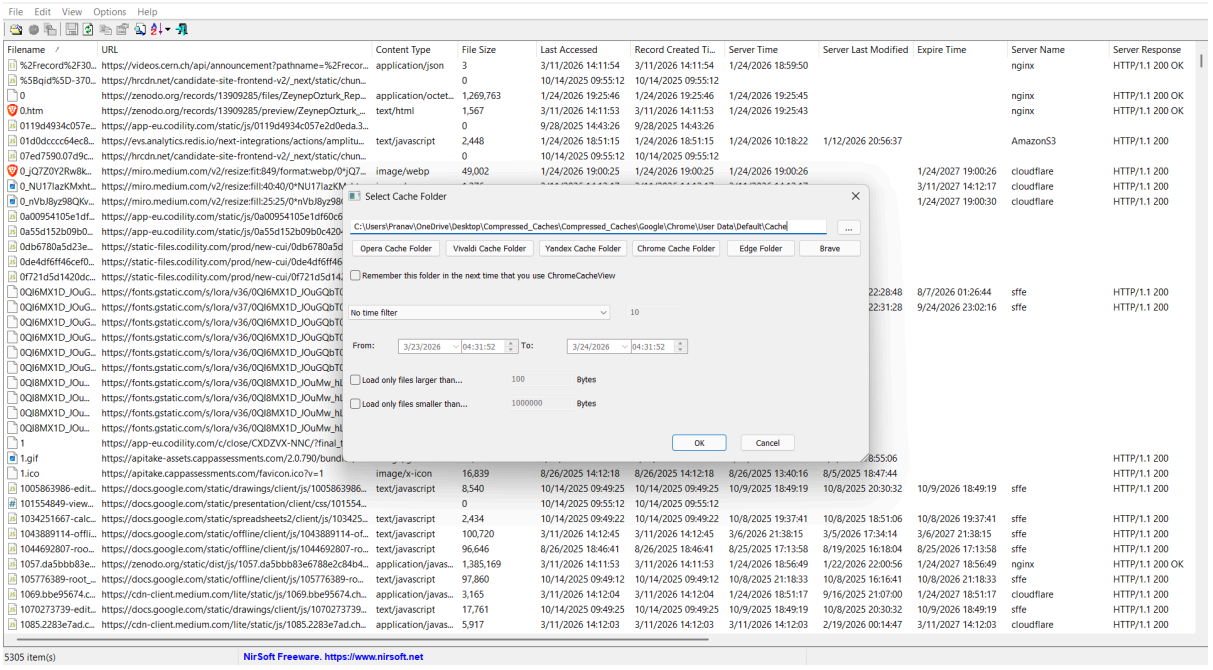
File Edit View Options Help

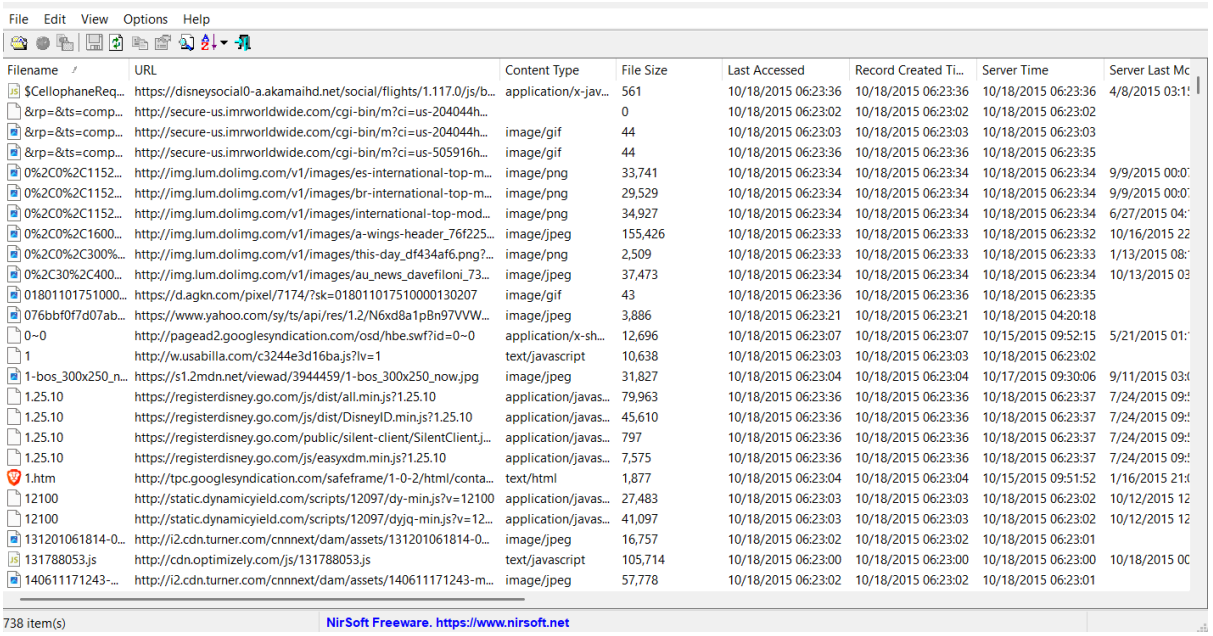
Filename	URL	Content Type	File Size	Last Accessed	Record Created Ti...	Server Time	Server Last Modified	Expire Time	Server Name	Server Response
%2Frecord%2F30...	https://videos.cern.ch/api/announcement?pathname=%2Frecor...	application/json	3	3/11/2026 14:11:54	3/11/2026 14:11:54	1/24/2026 18:59:50			nginx	HTTP/1.1 200 OK
%5Bqid%5D-370...	https://hrcdn.net/candidate-site-frontend-v2/_next/static/chun...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0	https://zenodo.org/records/13909285/files/ZeynepOzturk_Rep...	application/octet...	1,269,763	1/24/2026 19:25:46	1/24/2026 19:25:46	1/24/2026 19:25:45			nginx	HTTP/1.1 200 OK
0.htm	https://zenodo.org/records/13909285/preview/ZeynepOzturk_...	text/html	1,567	3/11/2026 14:11:53	3/11/2026 14:11:53	1/24/2026 19:25:43			nginx	HTTP/1.1 200 OK
0119d4934c057e...	https://app-eu.codility.com/static/js/0119d4934c057e2d0eda.3...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
01d0dcccc64ec8...	https://evs.analytics.redis.io/next-integrations/actions/amplitu...	text/javascript	2,448	1/24/2026 18:51:15	1/24/2026 18:51:15	1/24/2026 10:18:22	1/12/2026 20:56:37		AmazonS3	HTTP/1.1 200
07ed7590.07d9c...	https://hrcdn.net/candidate-site-frontend-v2/_next/static/chun...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0_jQ7Z0Y2Rw8k...	https://miro.medium.com/v2/resize:fit:849/format:webp/0*jQ7...	image/webp	49,002	1/24/2026 19:00:25	1/24/2026 19:00:25	1/24/2026 19:00:26		1/24/2027 19:00:26	cloudflare	HTTP/1.1 200
0_NU17lazKMxht...	https://miro.medium.com/v2/resize:fill:40:40/0*NU17lazKMxht...	image/png	1,276	3/11/2026 14:12:17	3/11/2026 14:12:17	3/11/2026 14:12:17		3/11/2027 14:12:17	cloudflare	HTTP/1.1 200
0_nVbJ8yz98QKv...	https://miro.medium.com/v2/resize:fill:25:25/0*nVbJ8yz98QKvd...	image/png	970	1/24/2026 19:00:29	1/24/2026 19:00:29	1/24/2026 19:00:30		1/24/2027 19:00:30	cloudflare	HTTP/1.1 200
0a00954105e1df...	https://app-eu.codility.com/static/js/0a00954105e1df60c6e8.5...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
0a55d152b09b0...	https://app-eu.codility.com/static/js/0a55d152b09b0c420457.7...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
0db6780a5d23e...	https://static-files.codility.com/prod/new-cui/0db6780a5d23e...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
0de4df6ff46cef0...	https://static-files.codility.com/prod/new-cui/0de4df6ff46cef0...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
0f721d5d1420dc...	https://static-files.codility.com/prod/new-cui/0f721d5d1420dc...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
0QI6MX1D_JOuG...	https://fonts.gstatic.com/s/lora/v36/0QI6MX1D_JOuGQbT0gvTJ...	font/woff2	1,420	8/26/2025 13:42:47	8/26/2025 13:42:47	8/7/2025 01:26:44	5/28/2025 22:28:48	8/7/2026 01:26:44	sffe	HTTP/1.1 200
0QI6MX1D_JOuG...	https://fonts.gstatic.com/s/lora/v37/0QI6MX1D_JOuGQbT0gvTJ...	font/woff2	1,420	10/14/2025 09:50:17	10/14/2025 09:50:17	9/24/2025 23:02:16	9/10/2025 22:31:28	9/24/2026 23:02:16	sffe	HTTP/1.1 200
0QI6MX1D_JOuG...	https://fonts.gstatic.com/s/lora/v36/0QI6MX1D_JOuGQbT0gvTJ...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI6MX1D_JOuG...	https://fonts.gstatic.com/s/lora/v36/0QI6MX1D_JOuGQbT0gvTJ...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI6MX1D_JOuG...	https://fonts.gstatic.com/s/lora/v36/0QI6MX1D_JOuGQbT0gvTJ...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI6MX1D_JOuG...	https://fonts.gstatic.com/s/lora/v36/0QI6MX1D_JOuGQbT0gvTJ...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v36/0QI8MX1D_JOuMw_hLdO6...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v36/0QI8MX1D_JOuMw_hLdO6...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v36/0QI8MX1D_JOuMw_hLdO6...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
0QI8MX1D_JOu...	https://fonts.gstatic.com/s/lora/v36/0QI8MX1D_JOuMw_hLdO6...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
1	https://app-eu.codility.com/c/close/CXDZVX-NNC/?final_task_...		0	9/28/2025 14:43:26	9/28/2025 14:43:26					
1.gif	https://apitake-assets.cappassessments.com/2.0.790/bundles/i...	image/gif	10,044	8/26/2025 20:13:18	8/26/2025 20:13:18	8/26/2025 13:40:15	8/5/2025 18:55:06			HTTP/1.1 200
1.ico	https://apitake.cappassessments.com/favicon.ico?v=1	image/x-icon	16,839	8/26/2025 14:12:18	8/26/2025 14:12:18	8/26/2025 13:40:16	8/5/2025 18:47:44			HTTP/1.1 200
1005863986-edit...	https://docs.google.com/static/drawings/client/js/1005863986...	text/javascript	8,540	10/14/2025 09:49:25	10/14/2025 09:49:25	10/9/2025 18:49:19	10/8/2025 20:30:32	10/9/2026 18:49:19	sffe	HTTP/1.1 200
101554849-view...	https://docs.google.com/static/presentation/client/css/101554...		0	10/14/2025 09:55:12	10/14/2025 09:55:12					
1034251667-calc...	https://docs.google.com/static/spreadsheets2/client/js/103425...	text/javascript	2,434	10/14/2025 09:49:22	10/14/2025 09:49:22	10/8/2025 19:37:41	10/8/2025 18:51:06	10/8/2026 19:37:41	sffe	HTTP/1.1 200
1043889114-offli...	https://docs.google.com/static/offline/client/js/1043889114-of...	text/javascript	100,720	3/11/2026 14:12:45	3/11/2026 14:12:45	3/6/2026 21:38:15	3/5/2026 17:34:14	3/6/2027 21:38:15	sffe	HTTP/1.1 200
1044692807-roo...	https://docs.google.com/static/offline/client/js/1044692807-ro...	text/javascript	96,646	8/26/2025 18:46:41	8/26/2025 18:46:41	8/25/2025 17:13:58	8/19/2025 16:18:04	8/25/2026 17:13:58	sffe	HTTP/1.1 200
1057.da5bbb83e...	https://zenodo.org/static/dist/js/1057.da5bbb83e6788e2c84b4...	application/javas...	1,385,169	3/11/2026 14:11:53	3/11/2026 14:11:53	1/24/2026 18:56:49	1/22/2026 22:00:56	1/24/2027 18:56:49	nginx	HTTP/1.1 200 OK
105776389-root_...	https://docs.google.com/static/offline/client/js/105776389-ro...	text/javascript	97,860	10/14/2025 09:49:12	10/14/2025 09:49:12	10/8/2025 21:18:33	10/8/2025 16:16:41	10/8/2026 21:18:33	sffe	HTTP/1.1 200
1069.bbe95674.c...	https://cdn-client.medium.com/lite/static/js/1069.bbe95674.ch...	application/javas...	3,165	3/11/2026 14:12:04	3/11/2026 14:12:04	1/24/2026 18:51:17	9/16/2025 21:07:00	1/24/2027 18:51:17	cloudflare	HTTP/1.1 200
1070273739-edit...	https://docs.google.com/static/drawings/client/js/1070273739...	text/javascript	17,761	10/14/2025 09:49:25	10/14/2025 09:49:25	10/9/2025 18:49:19	10/8/2025 20:30:32	10/9/2026 18:49:19	sffe	HTTP/1.1 200
1085.2283e7ad.c...	https://cdn-client.medium.com/lite/static/js/1085.2283e7ad.ch...	application/javas...	5,917	3/11/2026 14:12:03	3/11/2026 14:12:03	3/11/2026 14:12:03	2/19/2026 00:14:47	3/11/2027 14:12:03	cloudflare	HTTP/1.1 200

5305 item(s) NirSoft Freeware. https://www.nirsoft.net

6) By default, when ChromeCacheView is launched, it will point to the cache stored on the local hard drive for the user account, which launched the application.

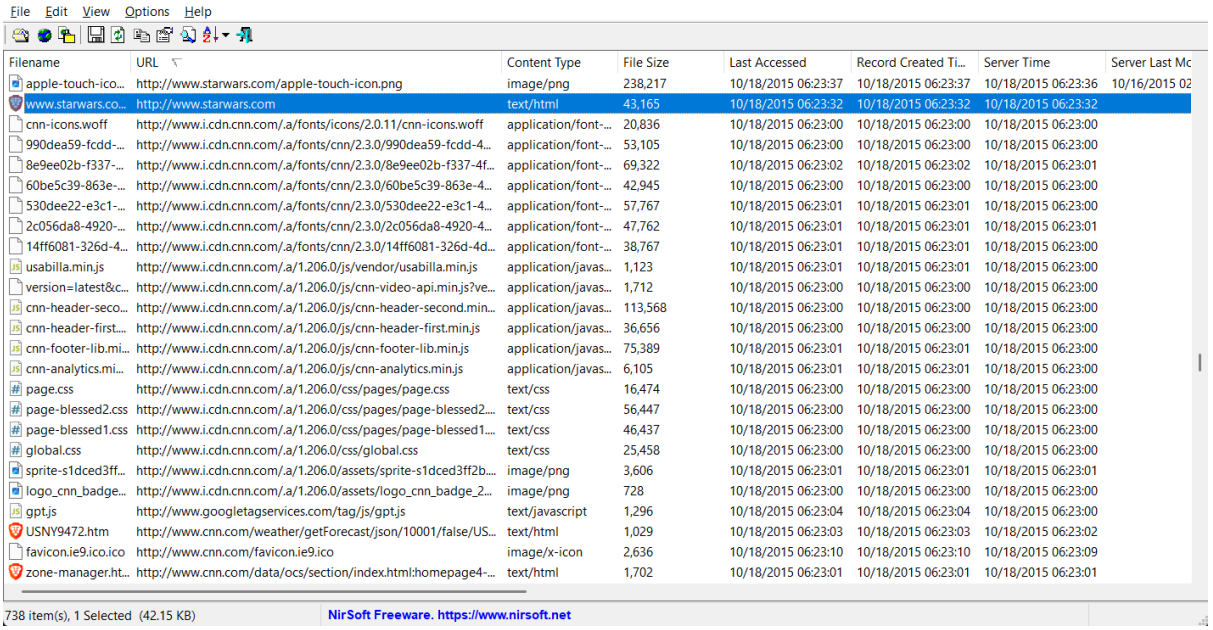
7) From the main menu select "File" and then choose "Select Cache Folder.' Browse to the following location in Desktop: \Desktop\Compressed_Caches\Google\Chrome\User Data\default\Cache, and click the "OK" button.





8) Click on the column labeled "URL" to sort the data alphabetically by URL.

Question 1. When was www.starwars.com visited?



File Edit View Options Help

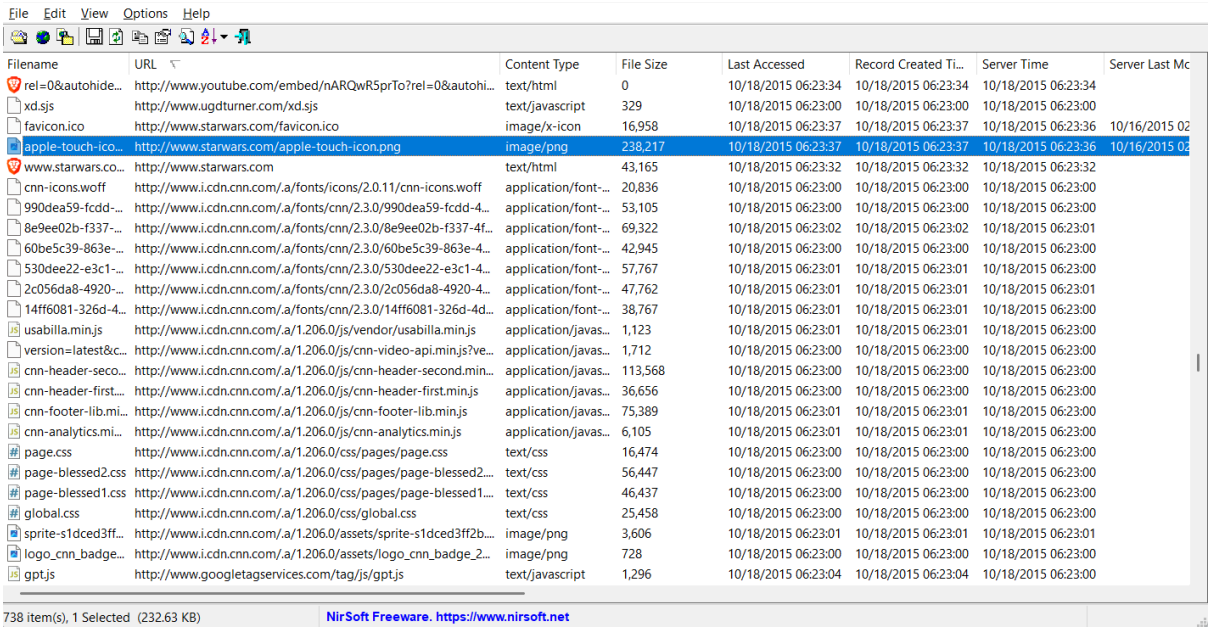
Filename	URL	Content Type	File Size	Last Accessed	Record Created Ti...	Server Time	Server Last Mc
apple-touch-ico...	http://www.starwars.com/apple-touch-icon.png	image/png	238,217	10/18/2015 06:23:37	10/18/2015 06:23:37	10/18/2015 06:23:36	10/16/2015 02
www.starwars.co...	http://www.starwars.com	text/html	43,165	10/18/2015 06:23:32	10/18/2015 06:23:32	10/18/2015 06:23:32	
cnn-icons.woff	http://www.i.cdn.cnn.com/.a/fonts/icons/2.0.11/cnn-icons.woff	application/font-...	20,836	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
990dea59-fcdd-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/990dea59-fcdd-4...	application/font-...	53,105	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
8e9ee02b-f337-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/8e9ee02b-f337-4f...	application/font-...	69,322	10/18/2015 06:23:02	10/18/2015 06:23:02	10/18/2015 06:23:01	
60be5c39-863e-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/60be5c39-863e-4...	application/font-...	42,945	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
530dee22-e3c1-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/530dee22-e3c1-4...	application/font-...	57,767	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
2c056da8-4920-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/2c056da8-4920-4...	application/font-...	47,762	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:01	
14ff6081-326d-4...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/14ff6081-326d-4d...	application/font-...	38,767	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
usabilla.min.js	http://www.i.cdn.cnn.com/.a/1.206.0/js/vendor/usabilla.min.js	application/javas...	1,123	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
version=latest&c...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-video-api.min.js?ve...	application/javas...	1,712	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
cnn-header-seco...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-header-second.min...	application/javas...	113,568	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
cnn-header-first....	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-header-first.min.js	application/javas...	36,656	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
cnn-footer-lib.mi...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-footer-lib.min.js	application/javas...	75,389	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
cnn-analytics.mi...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-analytics.min.js	application/javas...	6,105	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
page.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/pages/page.css	text/css	16,474	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
page-blessed2.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/pages/page-blessed2....	text/css	56,447	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
page-blessed1.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/pages/page-blessed1....	text/css	46,437	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
global.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/global.css	text/css	25,458	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
sprite-s1dced3ff...	http://www.i.cdn.cnn.com/.a/1.206.0/assets/sprite-s1dced3ff2b...	image/png	3,606	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:01	
logo_cnn_badge...	http://www.i.cdn.cnn.com/.a/1.206.0/assets/logo_cnn_badge_2...	image/png	728	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
gpt.js	http://www.googletagservices.com/tag/js/gpt.js	text/javascript	1,296	10/18/2015 06:23:04	10/18/2015 06:23:04	10/18/2015 06:23:00	
USNY9472.htm	http://www.cnn.com/weather/getForecast/json/10001/false/US...	text/html	1,029	10/18/2015 06:23:03	10/18/2015 06:23:03	10/18/2015 06:23:02	
favicon.ie9.ico.ico	http://www.cnn.com/favicon.ie9.ico	image/x-icon	2,636	10/18/2015 06:23:10	10/18/2015 06:23:10	10/18/2015 06:23:09	
zone-manager.ht...	http://www.cnn.com/data/ocs/section/index.html:homepage4-...	text/html	1,702	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:01	

738 item(s), 1 Selected (42.15 KB) NirSoft Freeware. https://www.nirsoft.net

Answer: As we can see above www.starwars.com was visited on 18th October 2015 at 06:23:32.

Explanation:

Question 2: When was the graphic "apple-touch-icon.png" last updated on the Star Wars web site?



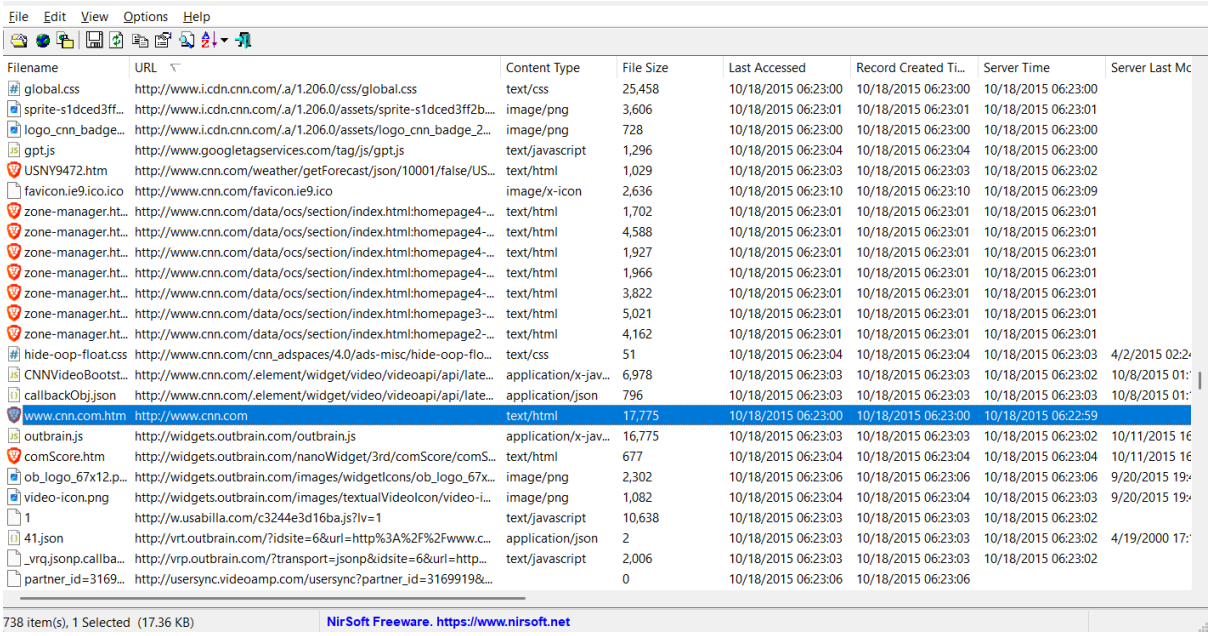
File Edit View Options Help

Filename	URL	Content Type	File Size	Last Accessed	Record Created Ti...	Server Time	Server Last Mc
rel=0&autohide...	http://www.youtube.com/embed/nARQwR5prTo?rel=0&autohi...	text/html	0	10/18/2015 06:23:34	10/18/2015 06:23:34	10/18/2015 06:23:34	
xd.sjs	http://www.ugdturner.com/xd.sjs	text/javascript	329	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
favicon.ico	http://www.starwars.com/favicon.ico	image/x-icon	16,958	10/18/2015 06:23:37	10/18/2015 06:23:37	10/18/2015 06:23:36	10/16/2015 02
apple-touch-ico...	http://www.starwars.com/apple-touch-icon.png	image/png	238,217	10/18/2015 06:23:37	10/18/2015 06:23:37	10/18/2015 06:23:36	10/16/2015 02
www.starwars.co...	http://www.starwars.com	text/html	43,165	10/18/2015 06:23:32	10/18/2015 06:23:32	10/18/2015 06:23:32	
cnn-icons.woff	http://www.i.cdn.cnn.com/.a/fonts/icons/2.0.11/cnn-icons.woff	application/font-...	20,836	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
990dea59-fcdd-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/990dea59-fcdd-4...	application/font-...	53,105	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
8e9ee02b-f337-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/8e9ee02b-f337-4f...	application/font-...	69,322	10/18/2015 06:23:02	10/18/2015 06:23:02	10/18/2015 06:23:01	
60be5c39-863e-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/60be5c39-863e-4...	application/font-...	42,945	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
530dee22-e3c1-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/530dee22-e3c1-4...	application/font-...	57,767	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
2c056da8-4920-...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/2c056da8-4920-4...	application/font-...	47,762	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:01	
14ff6081-326d-4...	http://www.i.cdn.cnn.com/.a/fonts/cnn/2.3.0/14ff6081-326d-4d...	application/font-...	38,767	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
usabilla.min.js	http://www.i.cdn.cnn.com/.a/1.206.0/js/vendor/usabilla.min.js	application/javas...	1,123	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
version=latest&c...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-video-api.min.js?ve...	application/javas...	1,712	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
cnn-header-seco...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-header-second.min...	application/javas...	113,568	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
cnn-header-first....	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-header-first.min.js	application/javas...	36,656	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
cnn-footer-lib.mi...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-footer-lib.min.js	application/javas...	75,389	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
cnn-analytics.mi...	http://www.i.cdn.cnn.com/.a/1.206.0/js/cnn-analytics.min.js	application/javas...	6,105	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:00	
page.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/pages/page.css	text/css	16,474	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
page-blessed2.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/pages/page-blessed2....	text/css	56,447	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
page-blessed1.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/pages/page-blessed1....	text/css	46,437	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
global.css	http://www.i.cdn.cnn.com/.a/1.206.0/css/global.css	text/css	25,458	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
sprite-s1dced3ff...	http://www.i.cdn.cnn.com/.a/1.206.0/assets/sprite-s1dced3ff2b...	image/png	3,606	10/18/2015 06:23:01	10/18/2015 06:23:01	10/18/2015 06:23:01	
logo_cnn_badge...	http://www.i.cdn.cnn.com/.a/1.206.0/assets/logo_cnn_badge_2...	image/png	728	10/18/2015 06:23:00	10/18/2015 06:23:00	10/18/2015 06:23:00	
gpt.js	http://www.googletagservices.com/tag/js/gpt.js	text/javascript	1,296	10/18/2015 06:23:04	10/18/2015 06:23:04	10/18/2015 06:23:00	

738 item(s), 1 Selected (232.63 KB) NirSoft Freeware. https://www.nirsoft.net

Answer: As per the record highlighted above, Apple-touch-icon.png was last updated on 16th October 2015 at 02:21:17 on the Star Wars website

Question3: When was www.cnn.com visited?



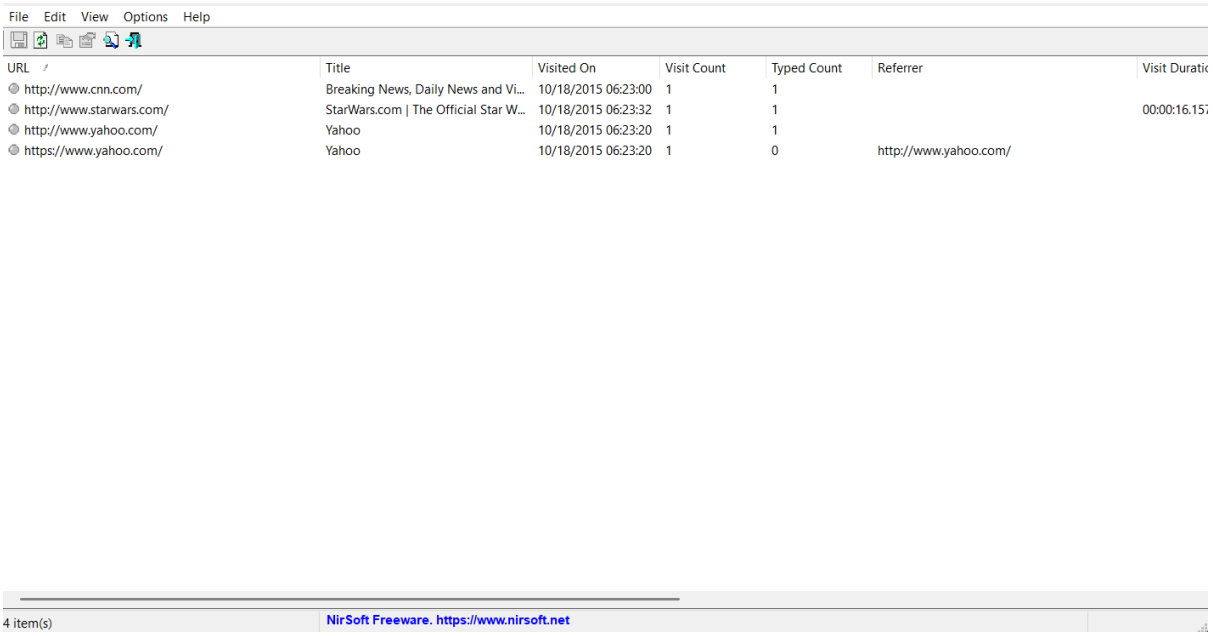
Answer: We can see that www.cnn.com was visited on 18th October 2015 at 06:23:00.

9) Launch ChromeHistoryView. By default, the application will point to the history stored on the local hard drive for the user account, which launched the application.

10) On main menu select "Options" and then select "Advanced Options."

11) Check the box next to "Use the following history file."

12) Browse to the following location in the Desktop: Desktop\Compressed Caches\Google\Chrome\UserData\Default\History, and click the "OK" button.



Question4: How many sites were visited?
Answer: Totally 4 websites were visited, namely http://www.cnn.com, http://www.starwars.com, http://www.yahoo.com and https://www.yahoo.com.

Activity 2: Analyzing Firefox Internet Cache and History

In this activity you will examine the cache maintained by Mozilla Firefox using NirSoft's MozillaCacheView and MozillaHistoryView.

Tools: Product: MozillaCacheView

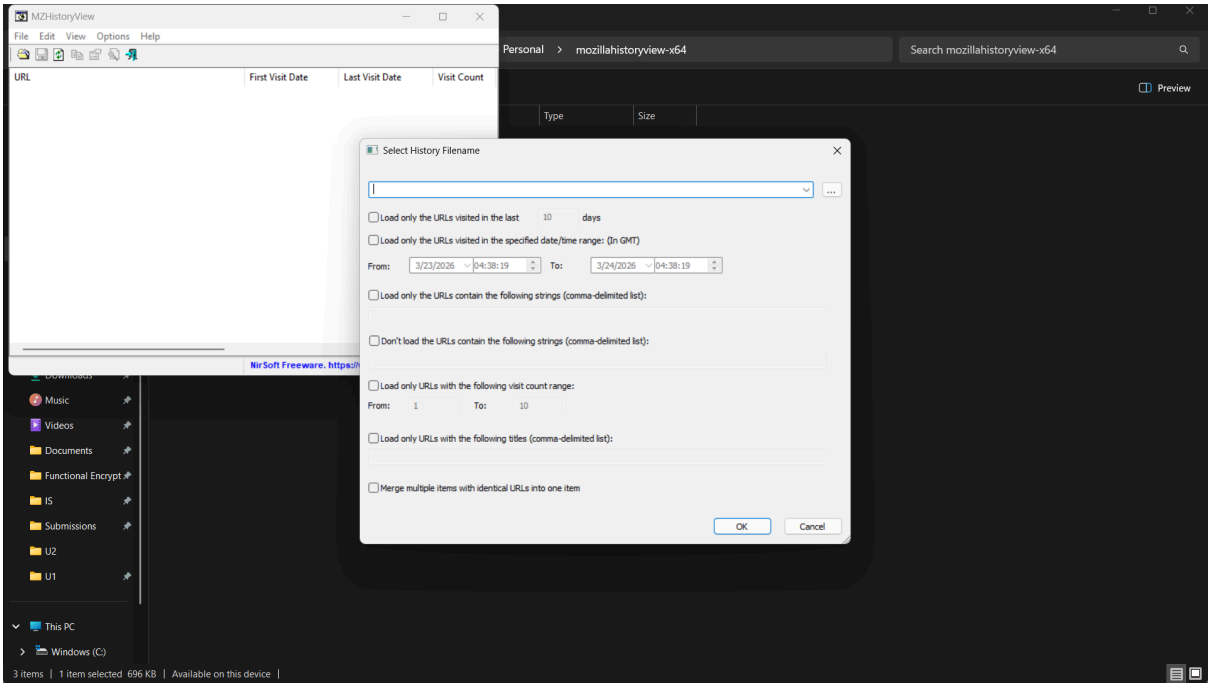
Manufacturer: NirSoft
Website:http://www.nirsoft.net/web_browser_tools.html

Product: MozillaHistoryView
Manufacturer: NirSoft
Website: http://www.nirsoft.net/web browser tools.html

13) Download MozillaCacheView. Extract the executable from the compressed file and place it on the desktop.
14) Download MozillaHistoryView. Extract the executable from the compressed file and place it on the desktop.
15) Download the file named "CompressedCaches.zip" given and extract the folders to your desktop. (Keep the folder structures intact.)
16) The cache for Mozilla Firefox was extracted from the following locations: C:\Users\admin\AppData\Local\Mozilla\Firefox\Profiles\profile.default\Cache

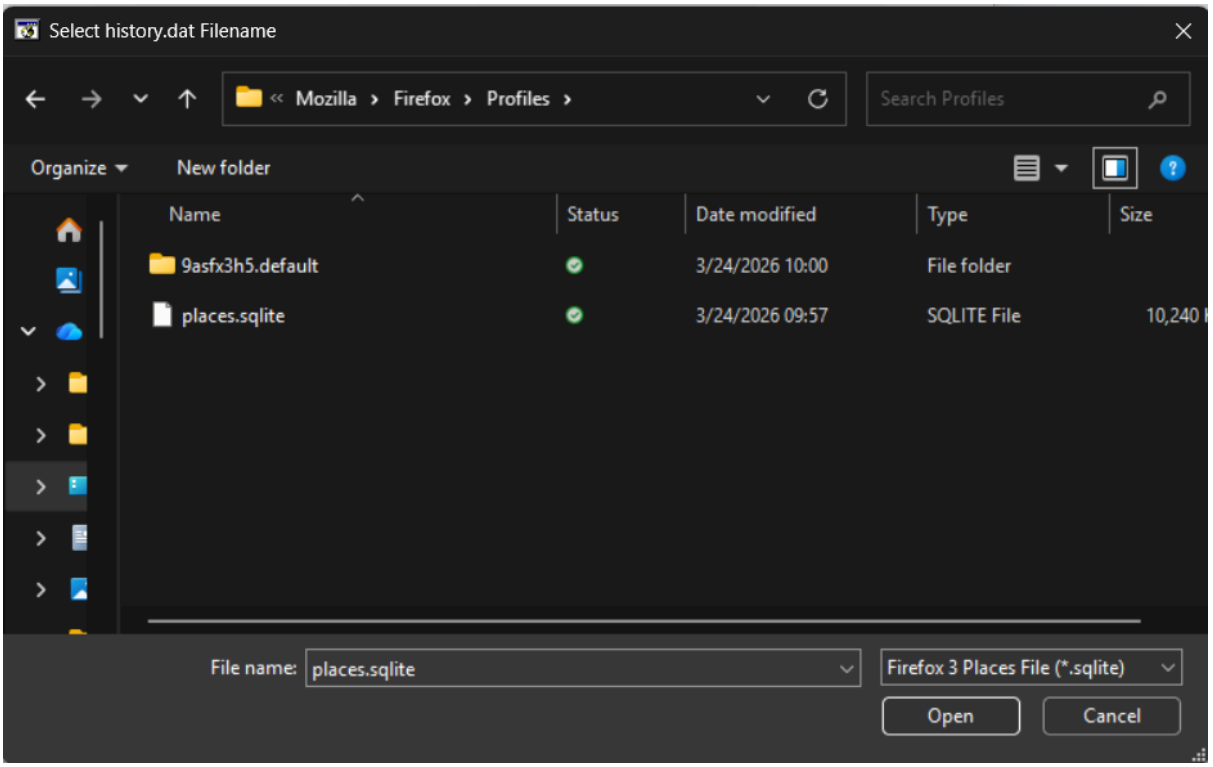
C:\Users\admin AppData\Roaming\Mozilla\Firefox\Profiles\profile.default places.sqlite\

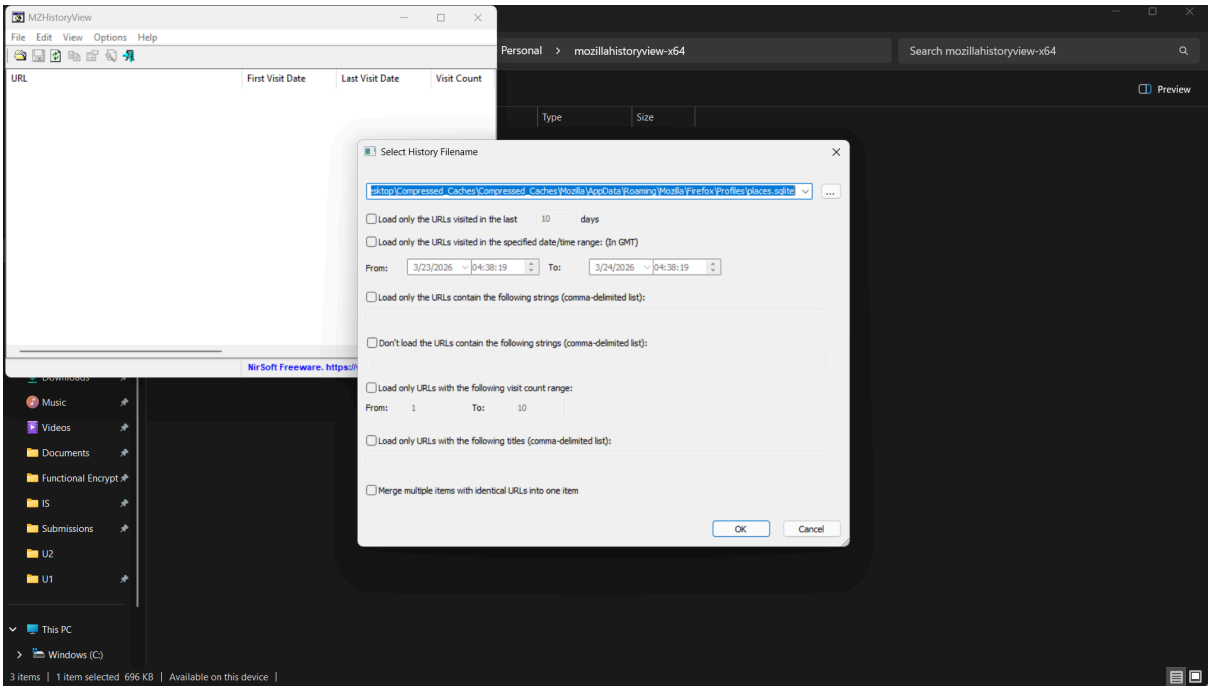
17)Launch MozillaHistoryView.

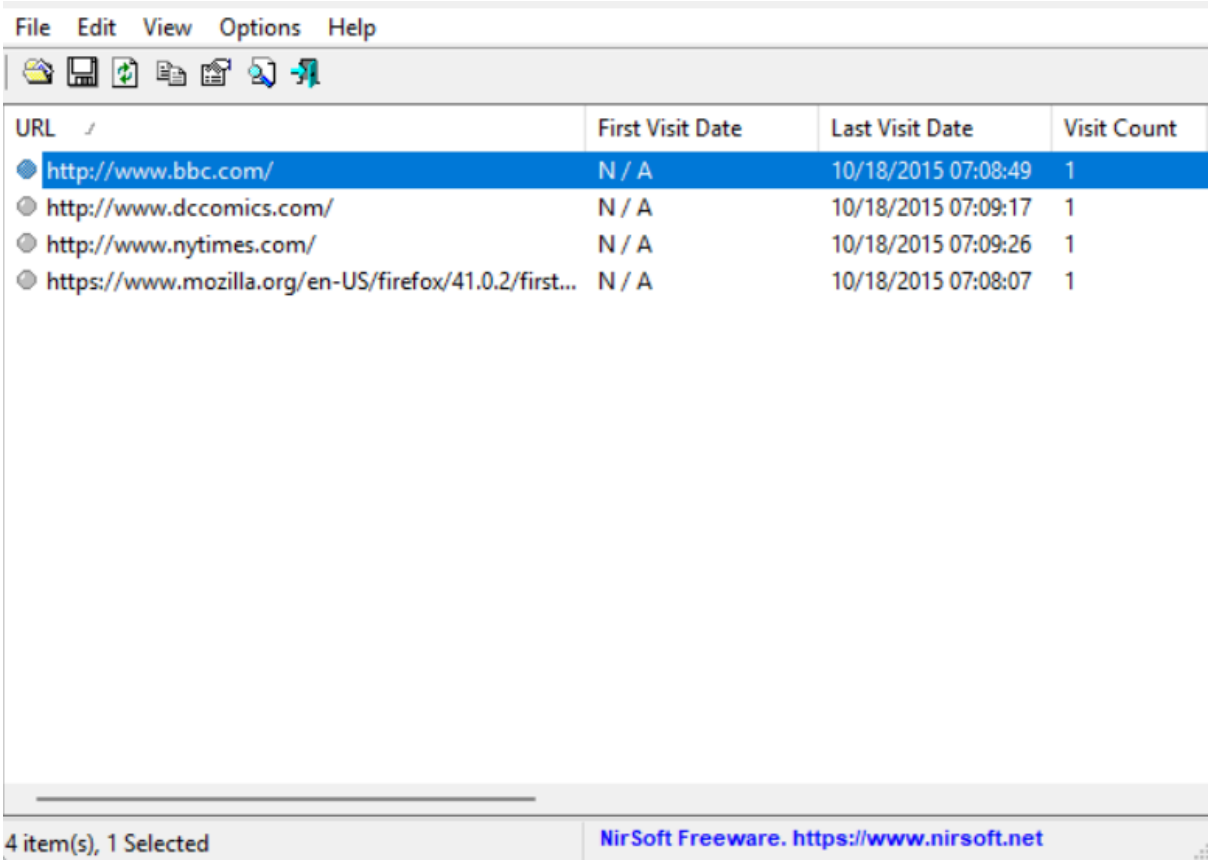


18)When MozillaHistoryView is launched, the user will be prompted to enter the path to the places.sqlite database. By default, the application will look inside the local user's profile. Browse to Desktop\Compressed Caches\Mozilla\AppData\Roaming\Mozilla\Firefox\Profiles\9asfx3h5.default\places.sqlite

(Note: It will be necessary to change the file type setting in the lower right corner of the screen using the pull-down menu. Change the option from "Mozilla history.dat to File (*.dat)" to "Firefox 3 places File (*.sqlite).) Click the "OK" button.

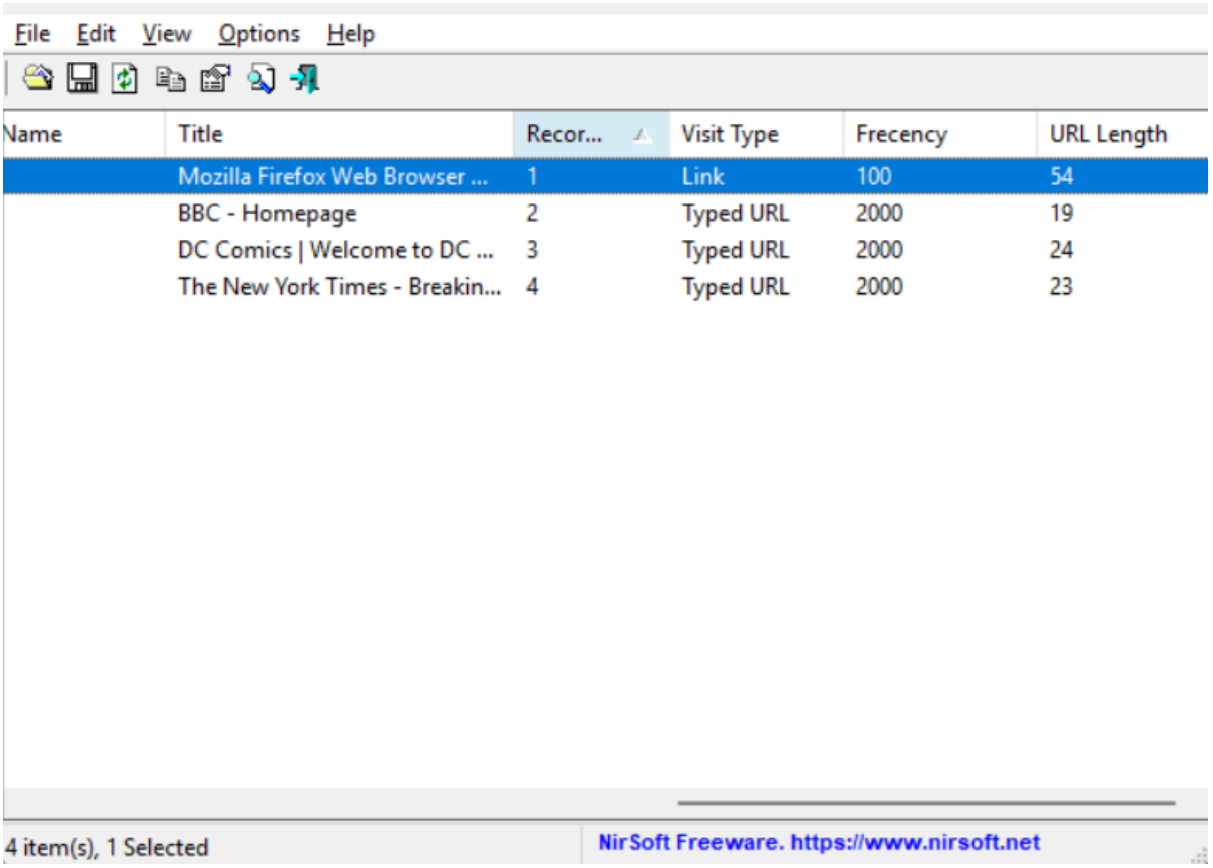

Select history.dat Filename
Mozilla › Firefox › Profiles ›
Search Profiles
Organize
New folder
Name
Status
Date modified
Type
Size
9asfx3h5.default
3/24/2026 10:00
File folder
places.sqlite
3/24/2026 09:57
SQLITE File
10,240
File name:
places.sqlite
Firefox 3 Places File (*.sqlite)
Open
Cancel


MZHistoryView
File Edit View Options Help
URL
First Visit Date
Last Visit Date
Visit Count
NirSoft Freeware. https://
Personal › mozillahistoryview-x64
Search mozillahistoryview-x64
Preview
Type
Size
Select History Filename
Load only the URLs visited in the last 10 days
Load only the URLs visited in the specified date/time range: (In GMT)
From: 3/23/2026 04:38:19 To: 3/24/2026 04:38:19
Load only the URLs contain the following strings (comma-delimited list):
Don't load the URLs contain the following strings (comma-delimited list):
Load only URLs with the following visit count range:
From: 1 To: 10
Load only URLs with the following titles (comma-delimited list):
Merge multiple items with identical URLs into one item
OK
Cancel
Music
Videos
Documents
Functional Encrypt
IS
Submissions
U2
U1
This PC
Windows (C:)
3 items | 1 item selected 696 KB | Available on this device



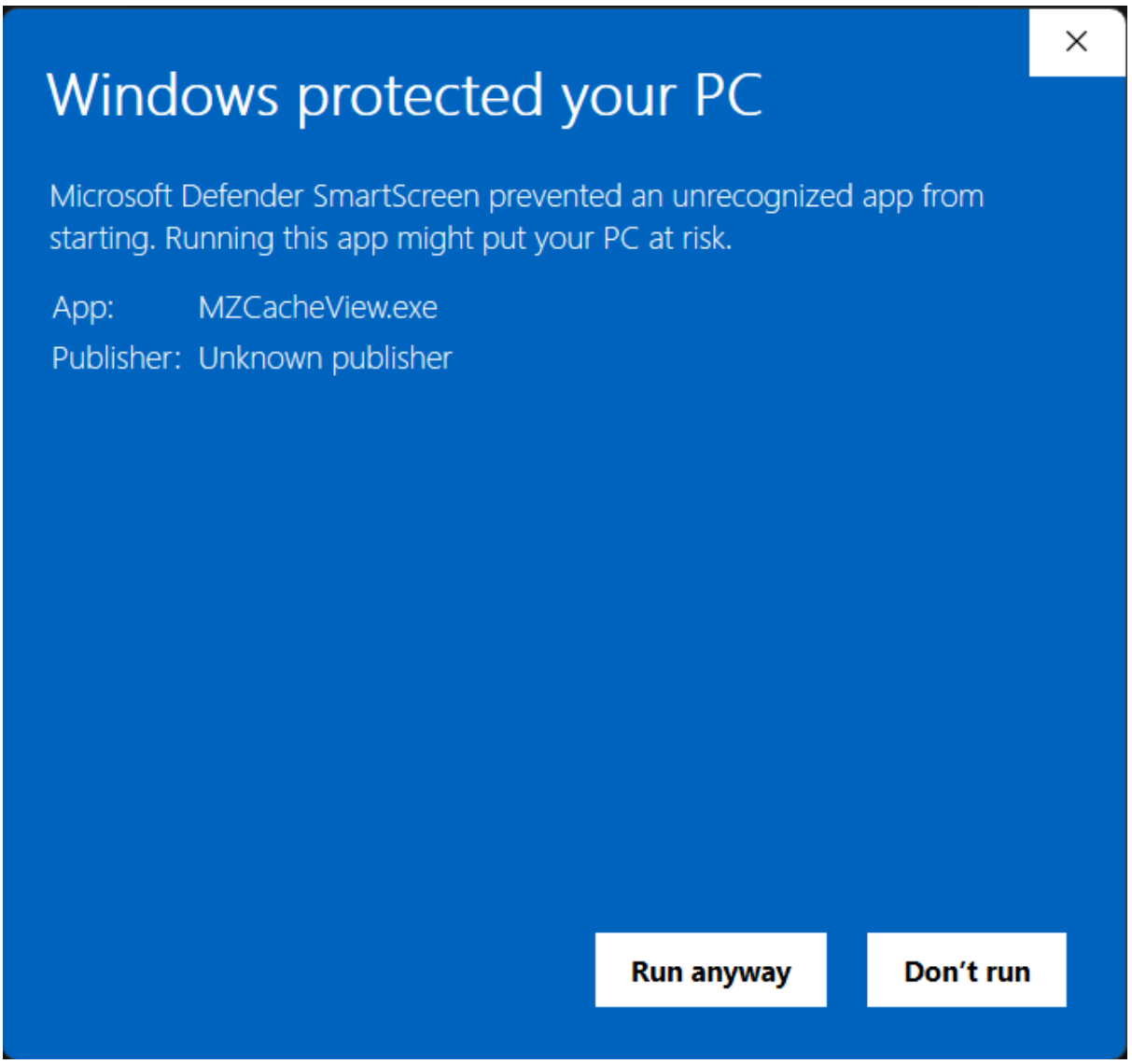
Question5: How many sites were visited?
Answer: Totally, 4 sites were visited, namely http://www.bbc.com, http://www.dccomics.com, http://www.nytimes.com and https://www.mozilla.org/en-US/firefox/41.0.2/firstrun/.

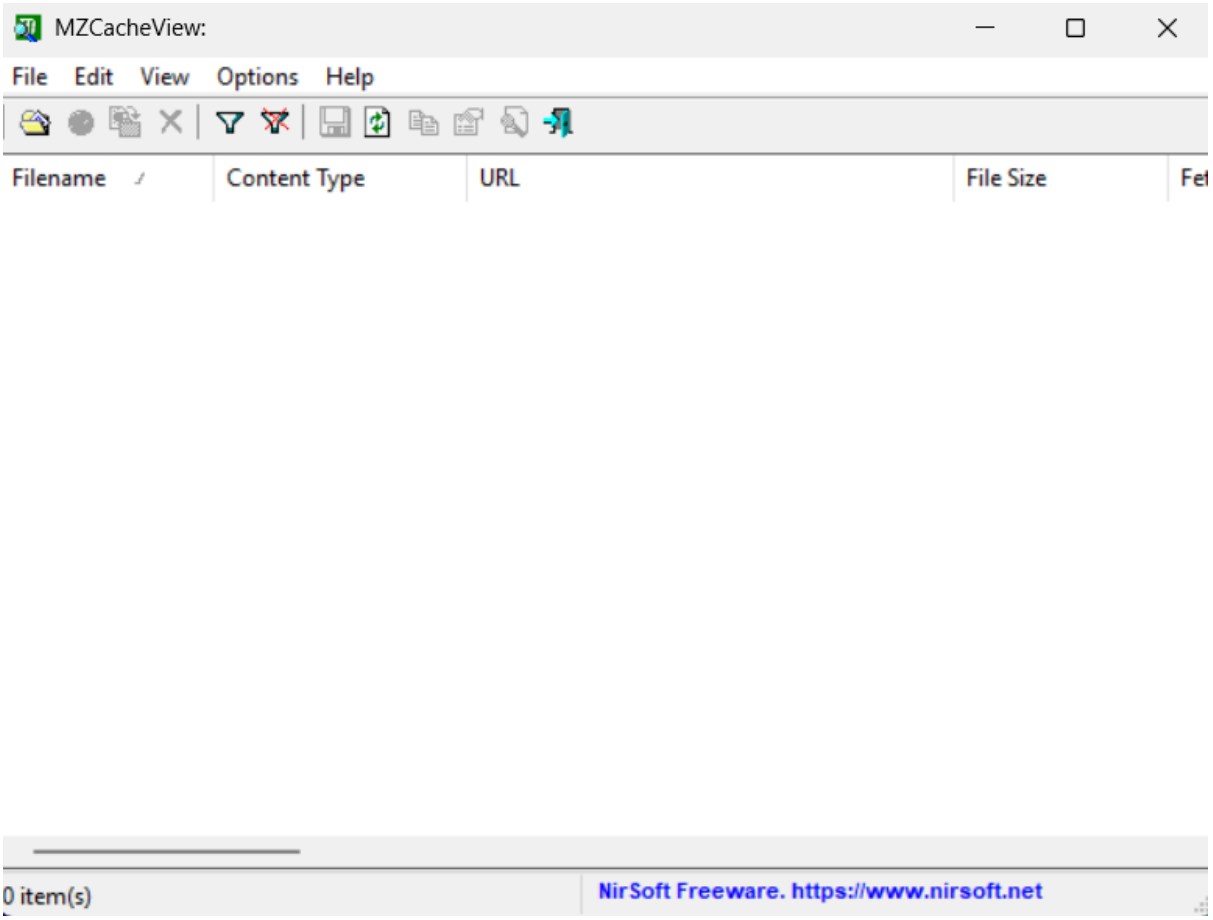
19) Scroll to the right site of the window and review the column named "Record Index".



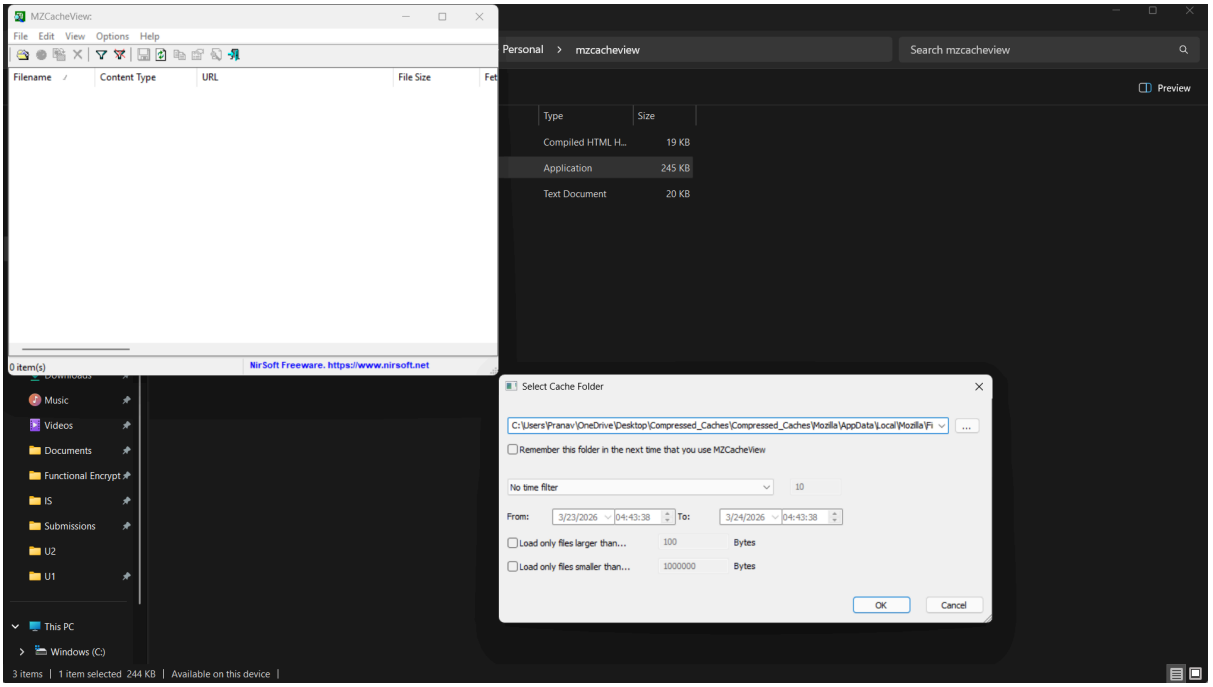
Question6: Which site was visited last?
Answer: As we can observe, the site last visited was http://www.nytimes.com.

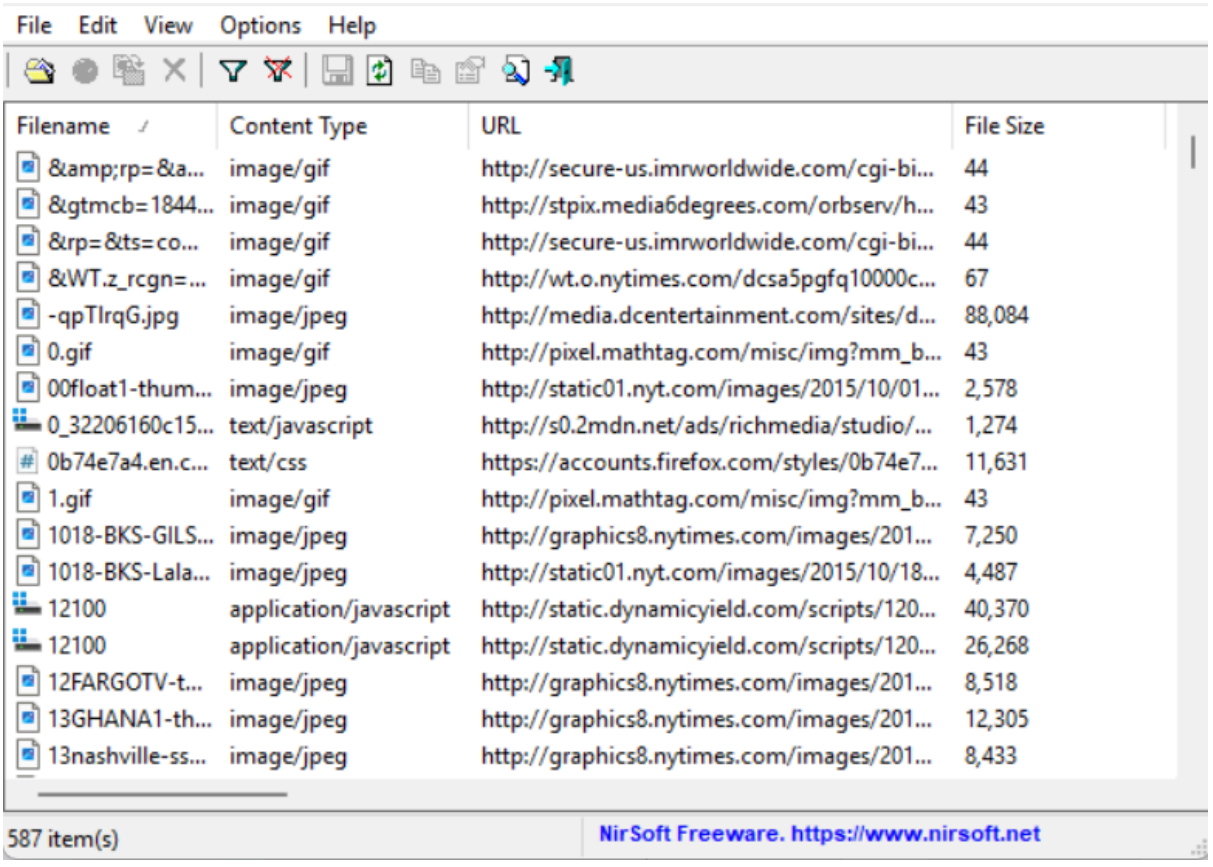
20)Launch MozillaCacheView.


Windows protected your PC
Microsoft Defender SmartScreen prevented an unrecognized app from starting. Running this app might put your PC at risk.
App: MZCacheView.exe
Publisher: Unknown publisher
Run anyway
Don't run



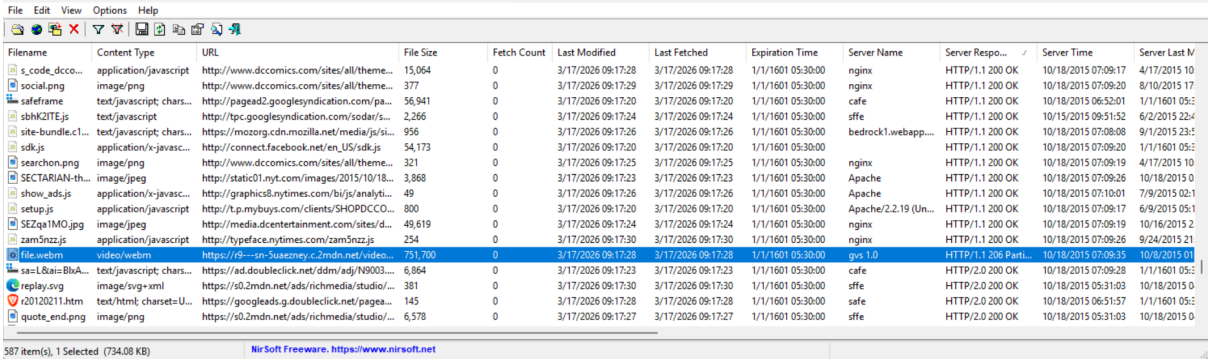
21) Redirect MozillaCacheView to the following directory in the Desktop: Desktop\Compressed Caches\Mozilla\AppData\Local\Mozilla\Firefox\Profiles\9asfx3h5.default\cache2 and press the "OK" button.





Question7: For the four visited web pages, how many items were cached?
Answer: For the four visited web pages, there are 587 items cached.

Question8: What is the only site to return an HTTP status code of 206, where the server returned only partial content?



Answer:
https://r9---sn-5uaezney.c.2mdn.net/videoplayback/id/d3dad3d01b717fff/itag/15/source/doubleclick/ratebypass/yes/mime/video%2Fwebm/acao/yes/ip/45.36.42.170/ipbits/0/expire/3589574359/sparams/acao,expire,id,ip,ipbits,itag,mime,mm,mn,ms,mv,nh,pl,ratebypass,source/signature/4081F3E7A5DF610DBE9CEA00854F5A4497D92A64.83F2D9236657F0C6D9F9D17B59090A5E364C629E/key/cms1/cms_redirect/yes/mm/30/mn/sn-5uaezney/ms/nxu/mt/1445132340/mv/m/nh/IgpwcjAyLmF0bDAxKgw3Mi4xNC4yMjEuODE/pl/15/file/file.webm was the only site that returned a status code of 206 (partial content).

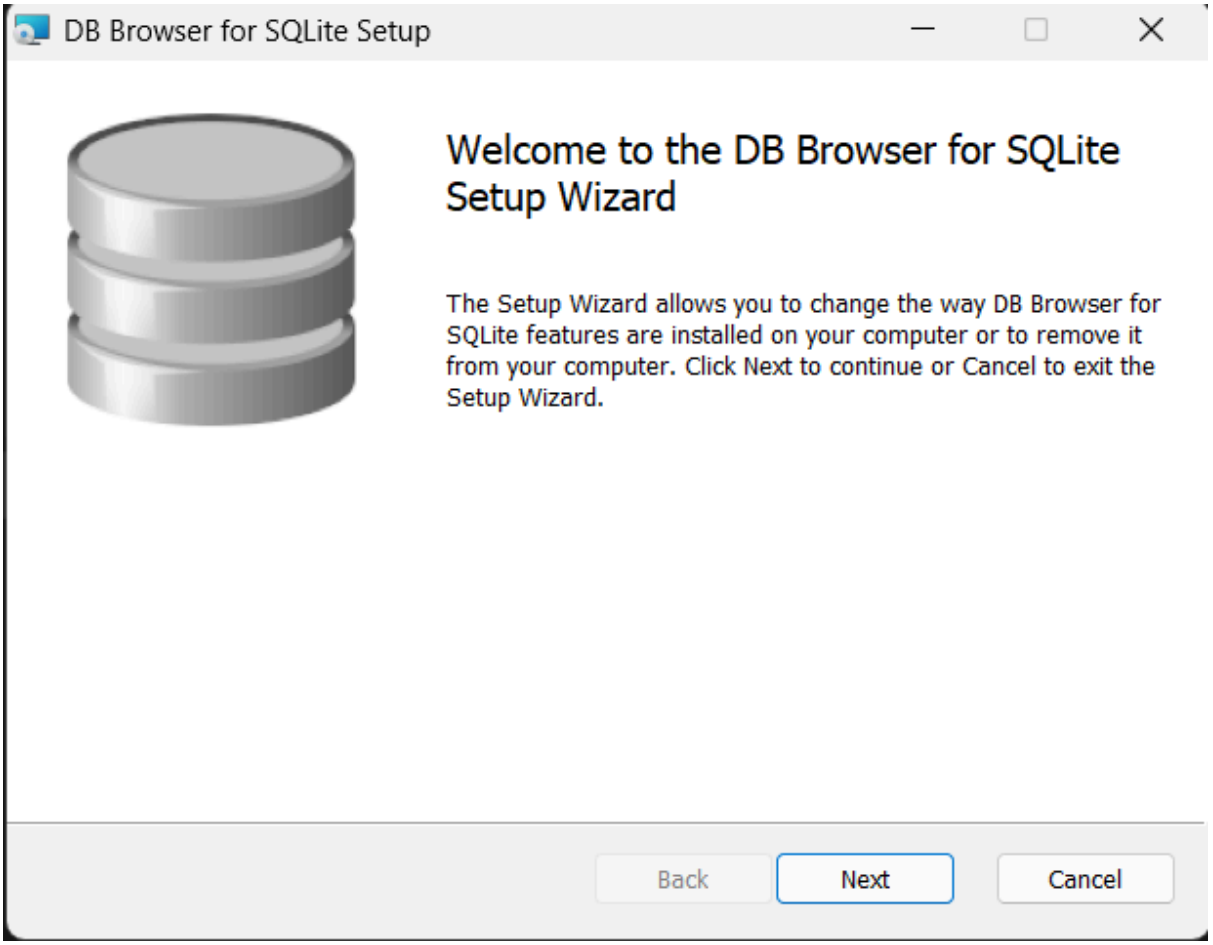
Activity 3: Analyzing Browser History using DB Browser for SQLite

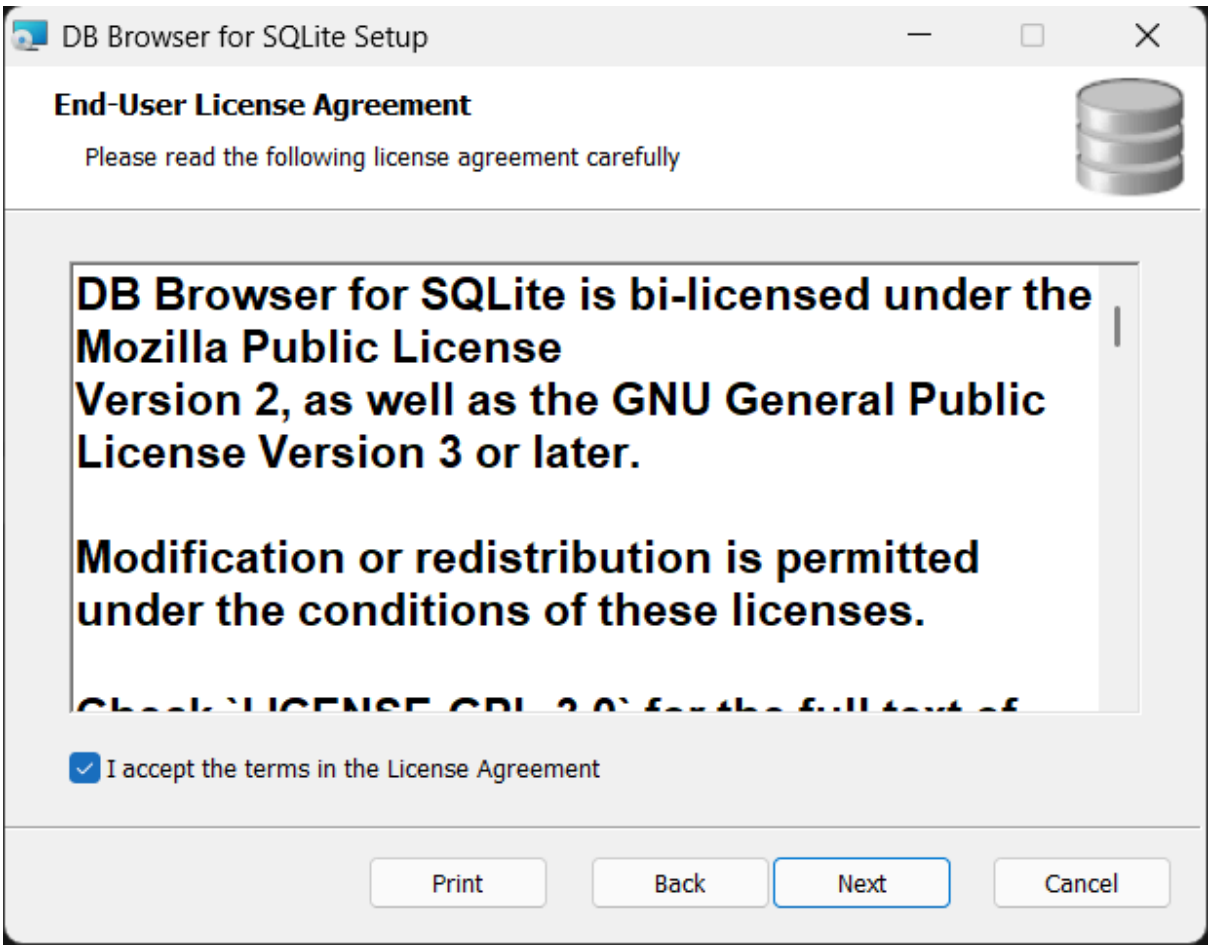
In this activity you will examine the browsing history maintained by Google Chrome using DB Browser for SQLite. Chrome stores browsing history in a SQLite database called History, which contains information about visited URLs, visit counts, and timestamps.

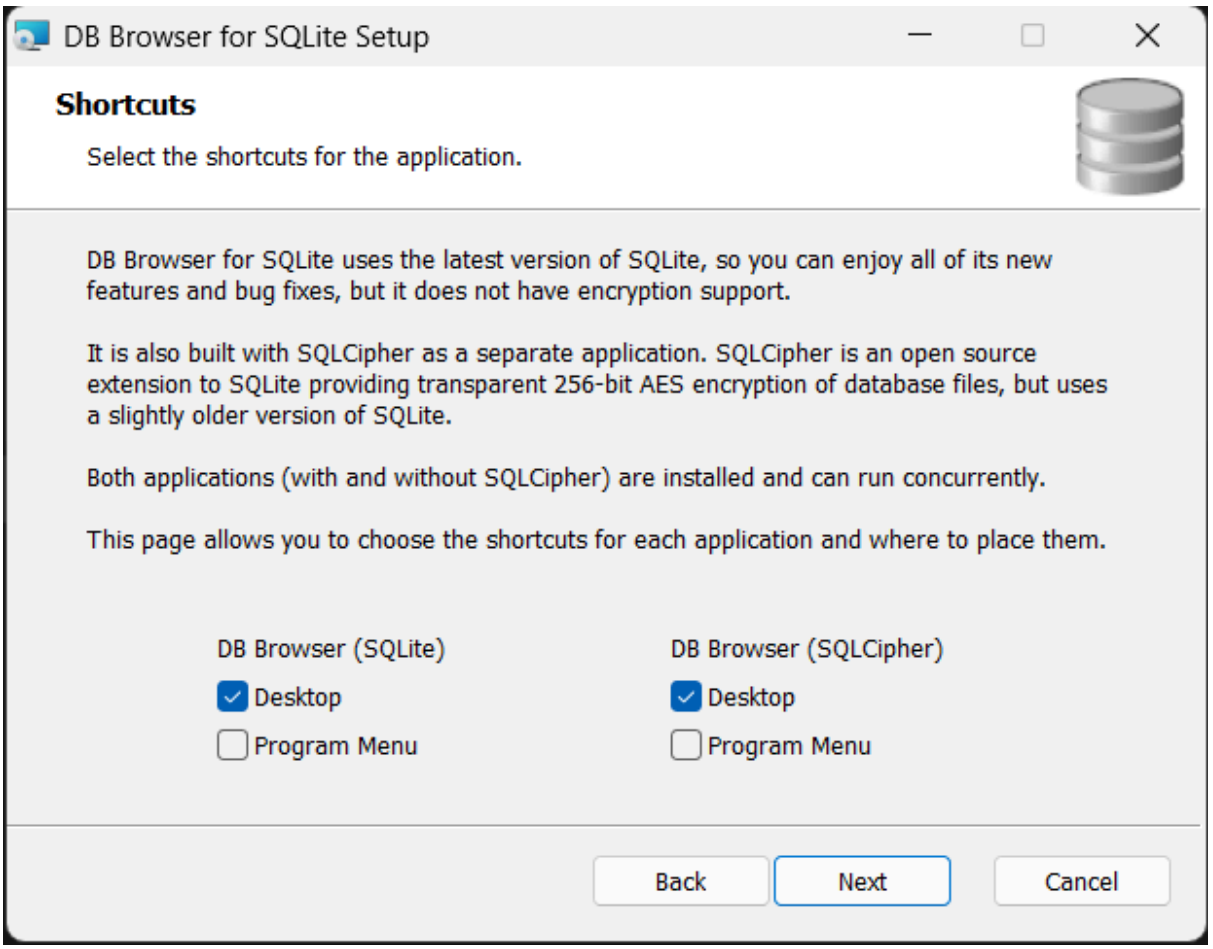
Tools: Product: DB Browser for SQLite

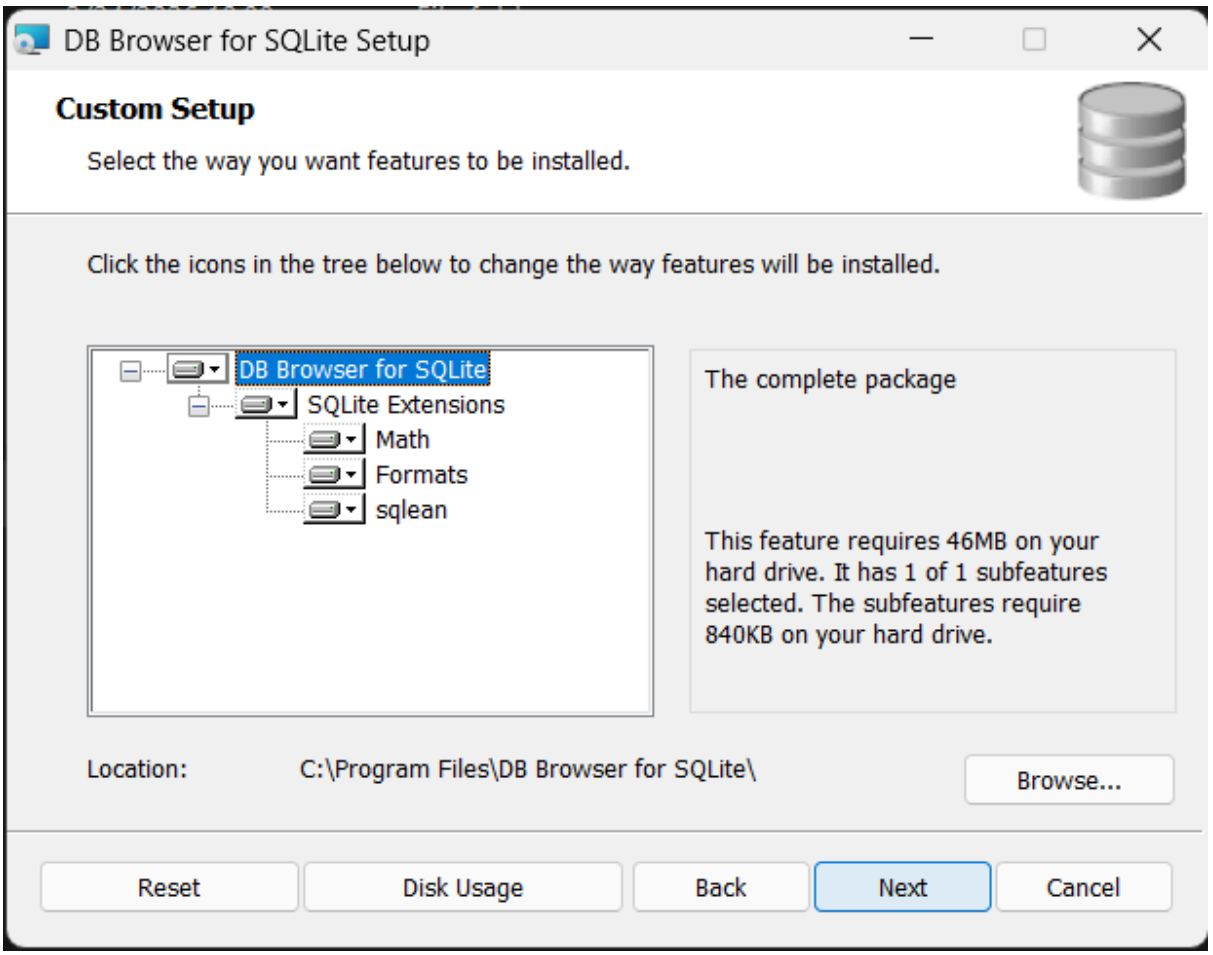
Manufacturer: SQLite Team
Website: https://sqlitebrowser.org/

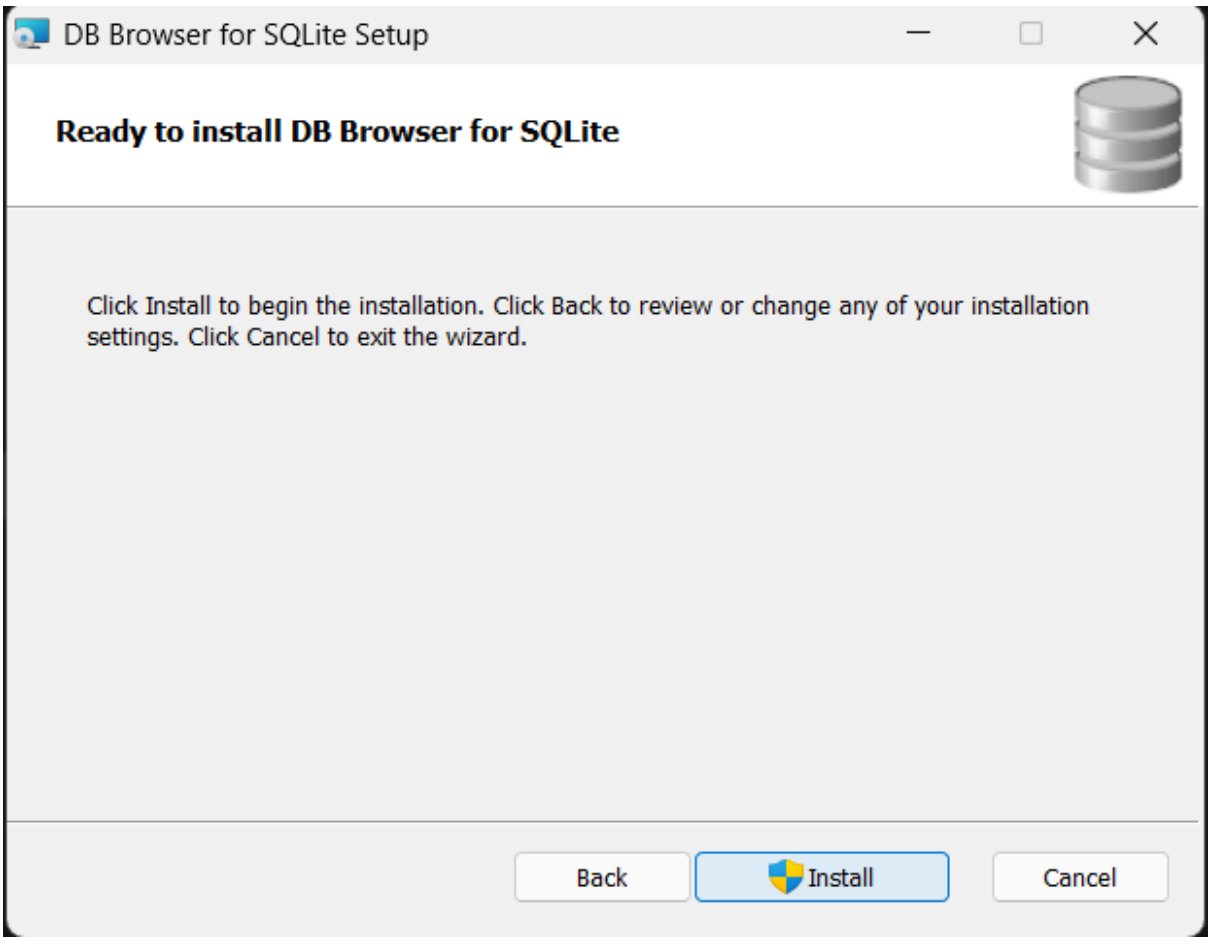
22)Download DB Browser for SQLite and install the application on the system.

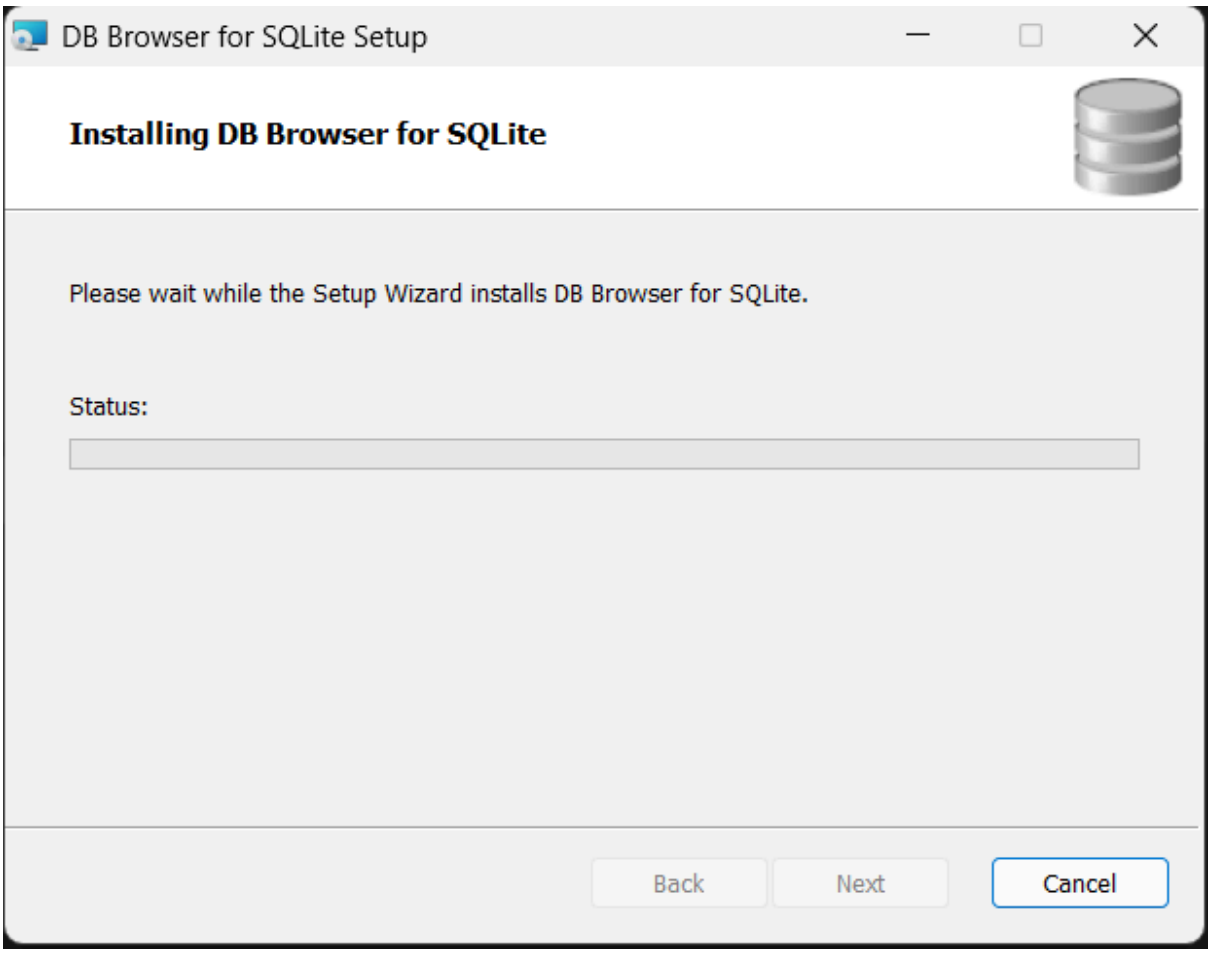

DB Browser for SQLite Setup
Welcome to the DB Browser for SQLite Setup Wizard
The Setup Wizard allows you to change the way DB Browser for SQLite features are installed on your computer or to remove it from your computer. Click Next to continue or Cancel to exit the Setup Wizard.
Back
Next
Cancel

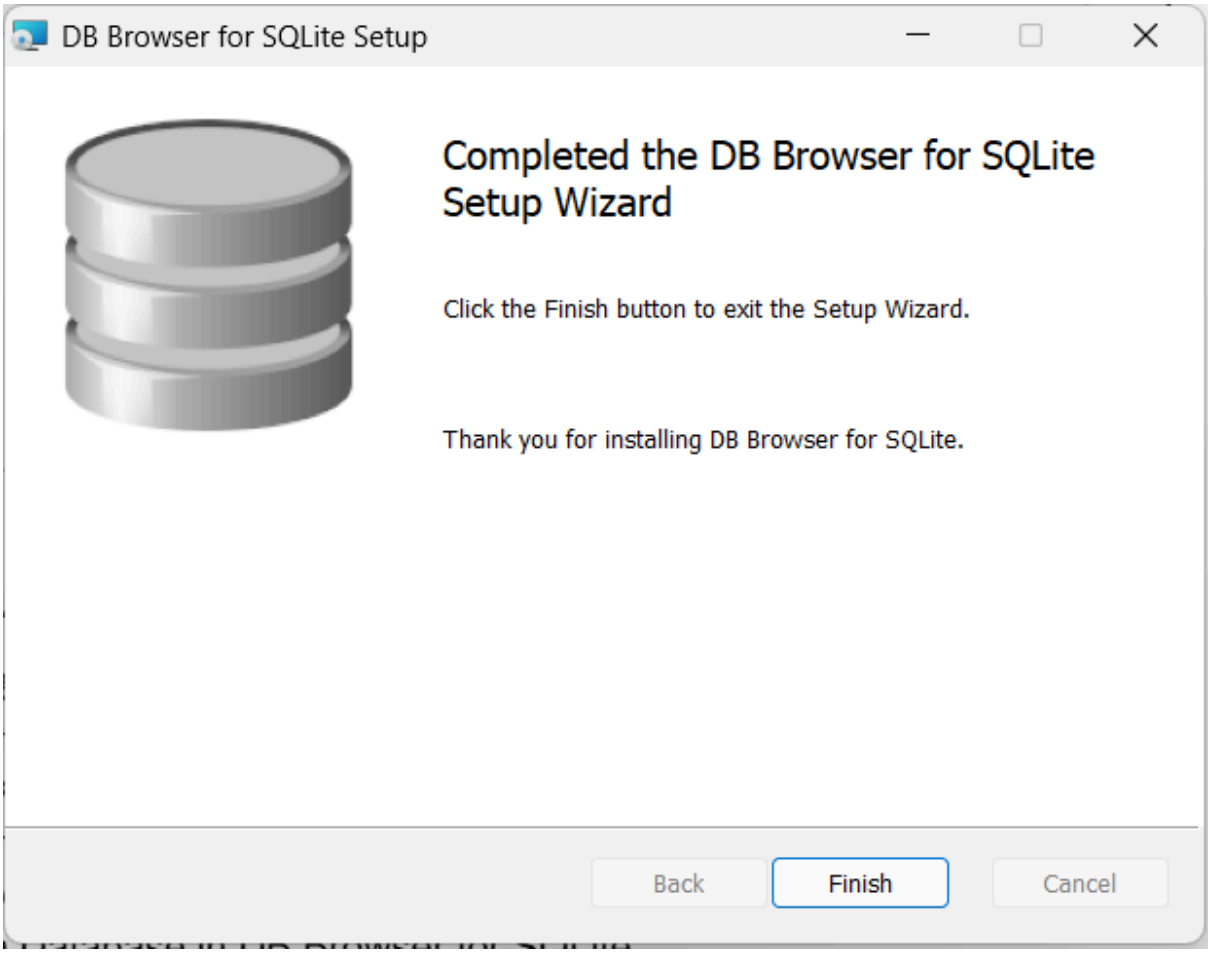

DB Browser for SQLite Setup
End-User License Agreement
Please read the following license agreement carefully
DB Browser for SQLite is bi-licensed under the Mozilla Public License
Version 2, as well as the GNU General Public License Version 3 or later.
Modification or redistribution is permitted under the conditions of these licenses.
I accept the terms in the License Agreement
Print
Back
Next
Cancel


DB Browser for SQLite Setup
Shortcuts
Select the shortcuts for the application.
DB Browser for SQLite uses the latest version of SQLite, so you can enjoy all of its new features and bug fixes, but it does not have encryption support.
It is also built with SQLCipher as a separate application. SQLCipher is an open source extension to SQLite providing transparent 256-bit AES encryption of database files, but uses a slightly older version of SQLite.
Both applications (with and without SQLCipher) are installed and can run concurrently.
This page allows you to choose the shortcuts for each application and where to place them.
DB Browser (SQLite)
Desktop
Program Menu
DB Browser (SQLCipher)
Desktop
Program Menu
Back
Next
Cancel

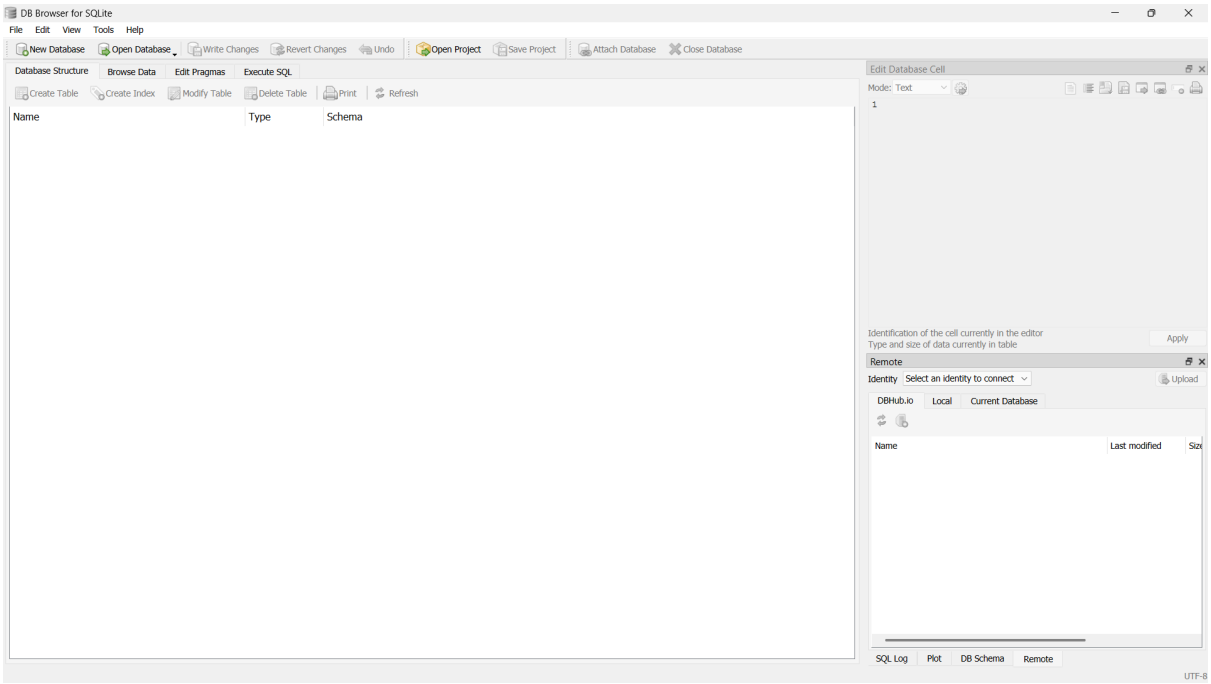

DB Browser for SQLite Setup
Custom Setup
Select the way you want features to be installed.
Click the icons in the tree below to change the way features will be installed.
DB Browser for SQLite
SQLite Extensions
Math
Formats
sqlean
The complete package
This feature requires 46MB on your hard drive. It has 1 of 1 subfeatures selected. The subfeatures require 840KB on your hard drive.
Location:
C:\Program Files\DB Browser for SQLite\
Browse...
Reset
Disk Usage
Back
Next
Cancel


DB Browser for SQLite Setup
Ready to install DB Browser for SQLite
Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.
Back
Install
Cancel

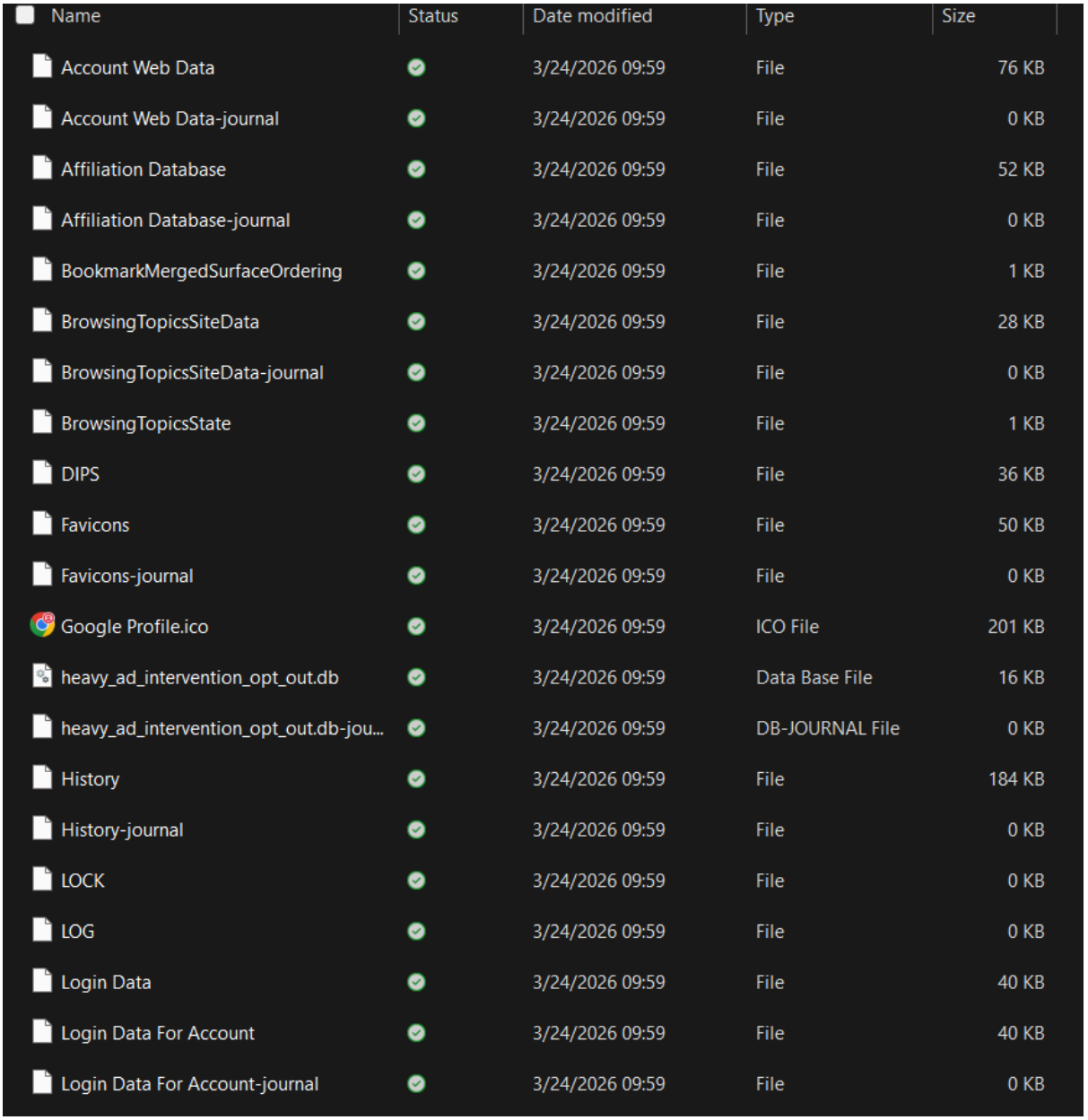

DB Browser for SQLite Setup
Installing DB Browser for SQLite
Please wait while the Setup Wizard installs DB Browser for SQLite.
Status:
Back
Next
Cancel



23)Launch DB Browser for SQLite.



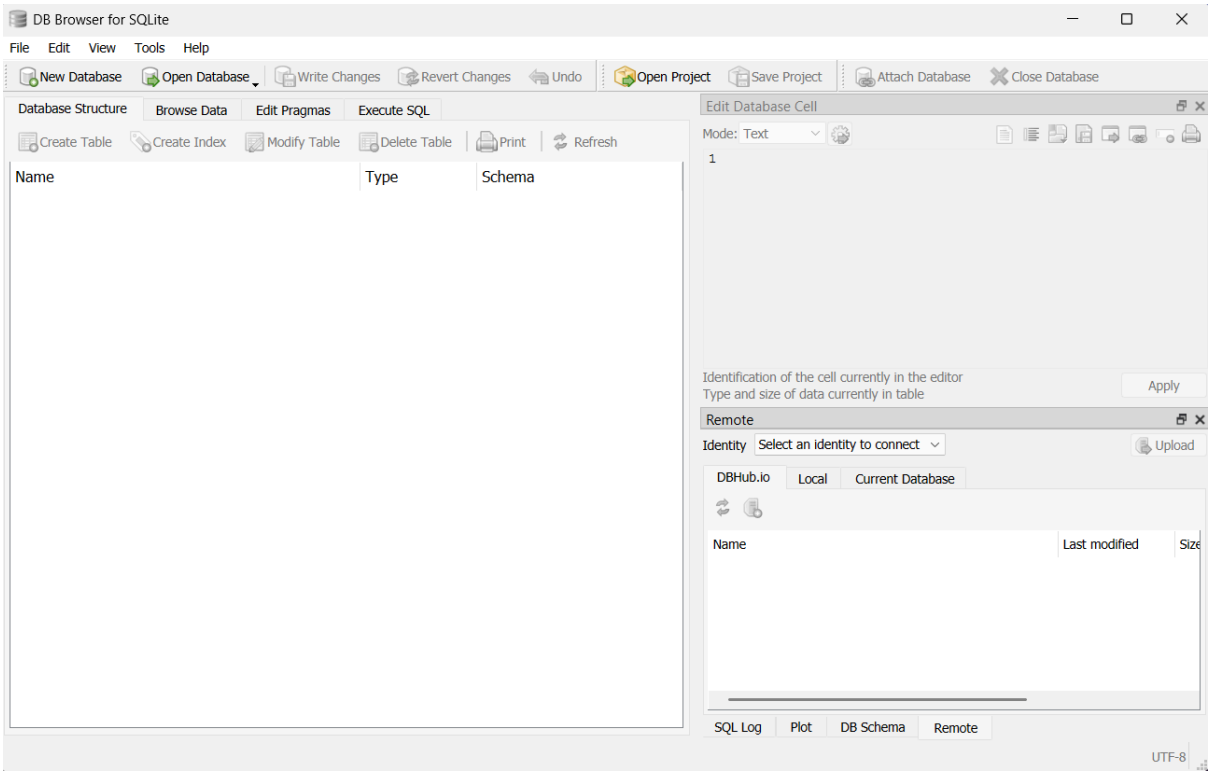
24)Download the file named DB analysis.zip given and extract the folders to your desktop. (Keep the folder structures intact.)



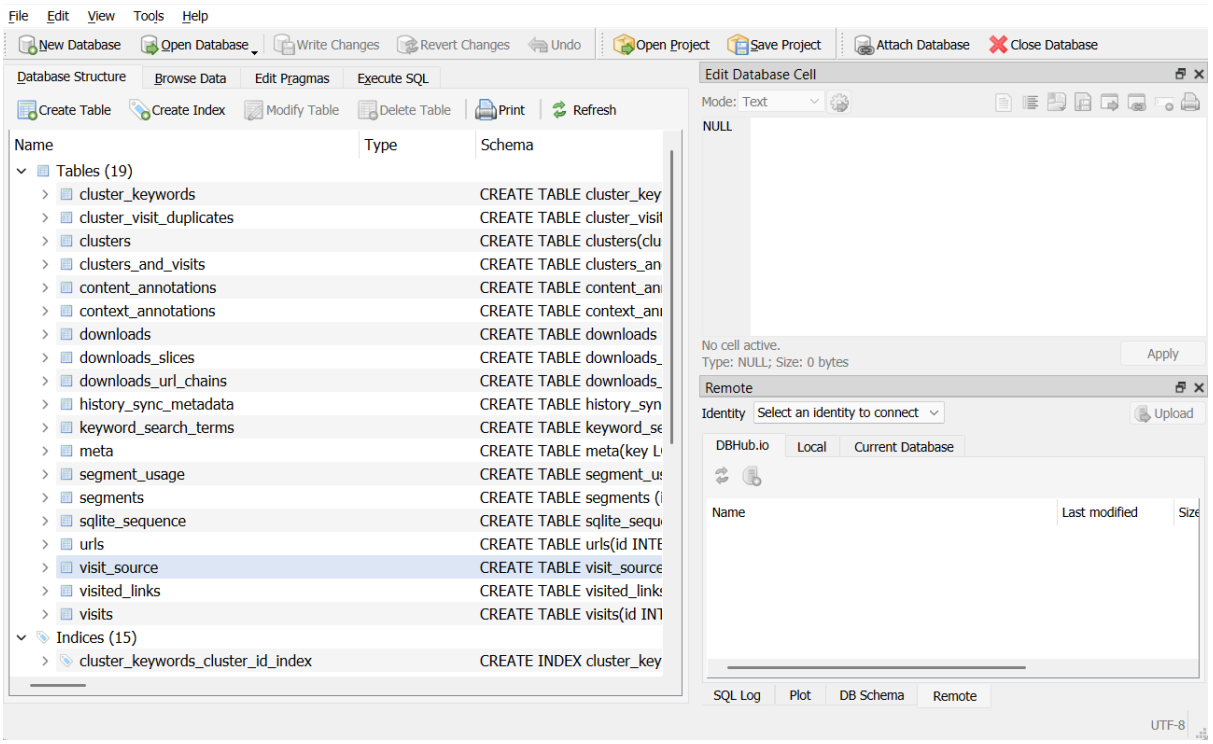
Name	Status	Date modified	Type	Size
Account Web Data	✓	3/24/2026 09:59	File	76 KB
Account Web Data-journal	✓	3/24/2026 09:59	File	0 KB
Affiliation Database	✓	3/24/2026 09:59	File	52 KB
Affiliation Database-journal	✓	3/24/2026 09:59	File	0 KB
BookmarkMergedSurfaceOrdering	✓	3/24/2026 09:59	File	1 KB
BrowsingTopicsSiteData	✓	3/24/2026 09:59	File	28 KB
BrowsingTopicsSiteData-journal	✓	3/24/2026 09:59	File	0 KB
BrowsingTopicsState	✓	3/24/2026 09:59	File	1 KB
DIPS	✓	3/24/2026 09:59	File	36 KB
Favicons	✓	3/24/2026 09:59	File	50 KB
Favicons-journal	✓	3/24/2026 09:59	File	0 KB
Google Profile.ico	✓	3/24/2026 09:59	ICO File	201 KB
heavy_ad_intervention_opt_out.db	✓	3/24/2026 09:59	Data Base File	16 KB
heavy_ad_intervention_opt_out.db-jou...	✓	3/24/2026 09:59	DB-JOURNAL File	0 KB
History	✓	3/24/2026 09:59	File	184 KB
History-journal	✓	3/24/2026 09:59	File	0 KB
LOCK	✓	3/24/2026 09:59	File	0 KB
LOG	✓	3/24/2026 09:59	File	0 KB
Login Data	✓	3/24/2026 09:59	File	40 KB
Login Data For Account	✓	3/24/2026 09:59	File	40 KB
Login Data For Account-journal	✓	3/24/2026 09:59	File	0 KB

25) The Chrome History database was extracted from the following location:
C:\Users\admin\AppData\Local\Google\Chrome\User Data\Default\History

26) Click Open Database in DB Browser for SQLite.

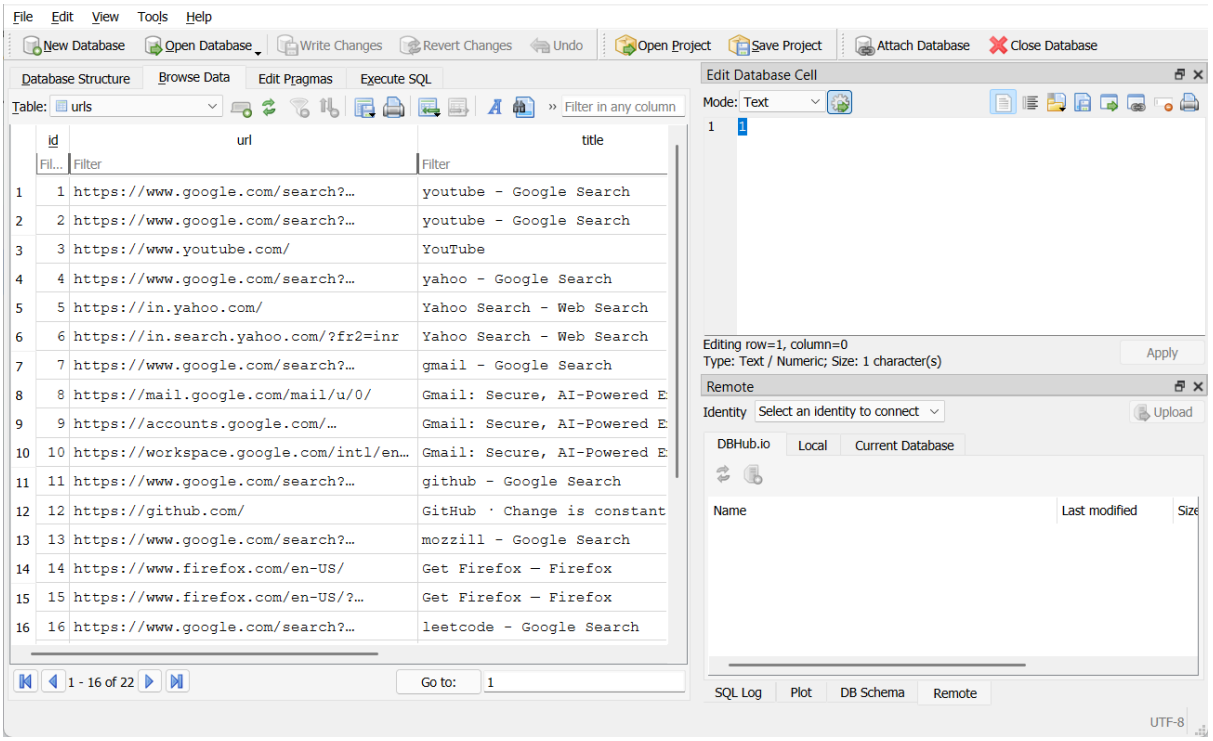


27)Browse to the following location (Make sure to select All files) and click the Open button.
\DB analysis\History



28)Click on the Browse Data tab.

29)From the Table drop-down menu select the table named urls.

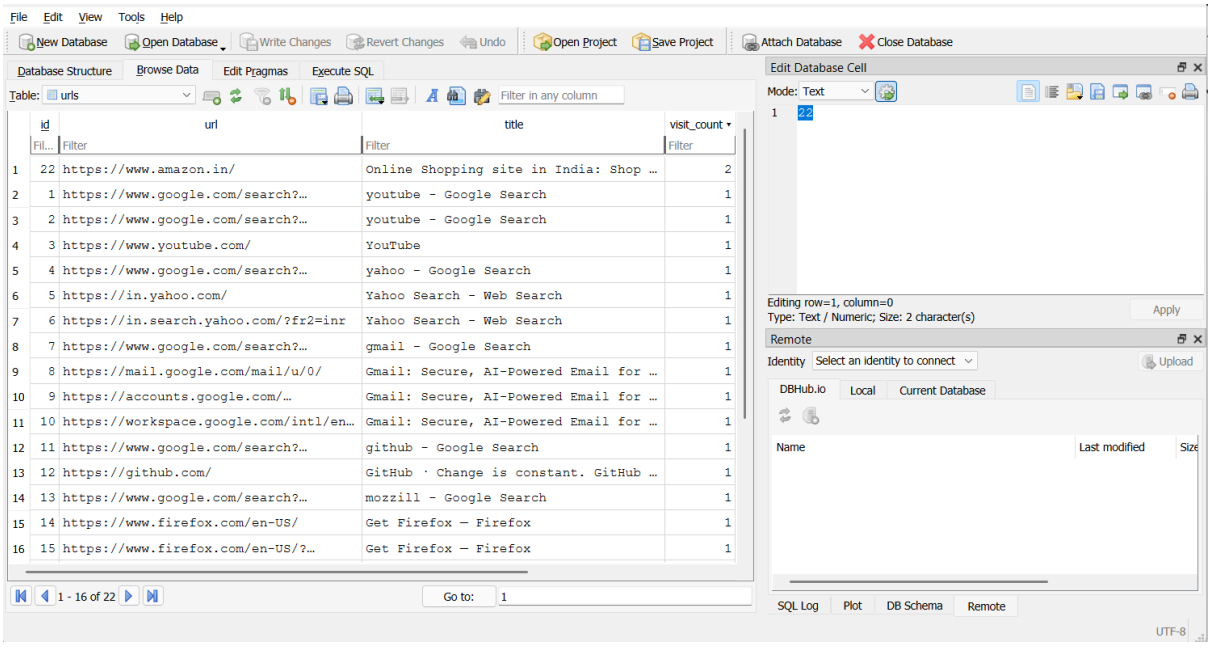


	id	url	title
1	1	https://www.google.com/search?…	youtube - Google Search
2	2	https://www.google.com/search?…	youtube - Google Search
3	3	https://www.youtube.com/	YouTube
4	4	https://www.google.com/search?…	yahoo - Google Search
5	5	https://in.yahoo.com/	Yahoo Search - Web Search
6	6	https://in.search.yahoo.com/?fr2=inr	Yahoo Search - Web Search
7	7	https://www.google.com/search?…	gmail - Google Search
8	8	https://mail.google.com/mail/u/0/	Gmail: Secure, AI-Powered E
9	9	https://accounts.google.com/…	Gmail: Secure, AI-Powered E
10	10	https://workspace.google.com/intl/en…	Gmail: Secure, AI-Powered E
11	11	https://www.google.com/search?…	github - Google Search
12	12	https://github.com/	GitHub · Change is constant
13	13	https://www.google.com/search?…	mozzill - Google Search
14	14	https://www.firefox.com/en-US/	Get Firefox — Firefox
15	15	https://www.firefox.com/en-US/?…	Get Firefox — Firefox
16	16	https://www.google.com/search?…	leetcode - Google Search

30)Review the list of visited URLs in the table.

Question9: How many URLs are present in the urls table?
Answer: There are 22 URLs in the urls table as we observe.

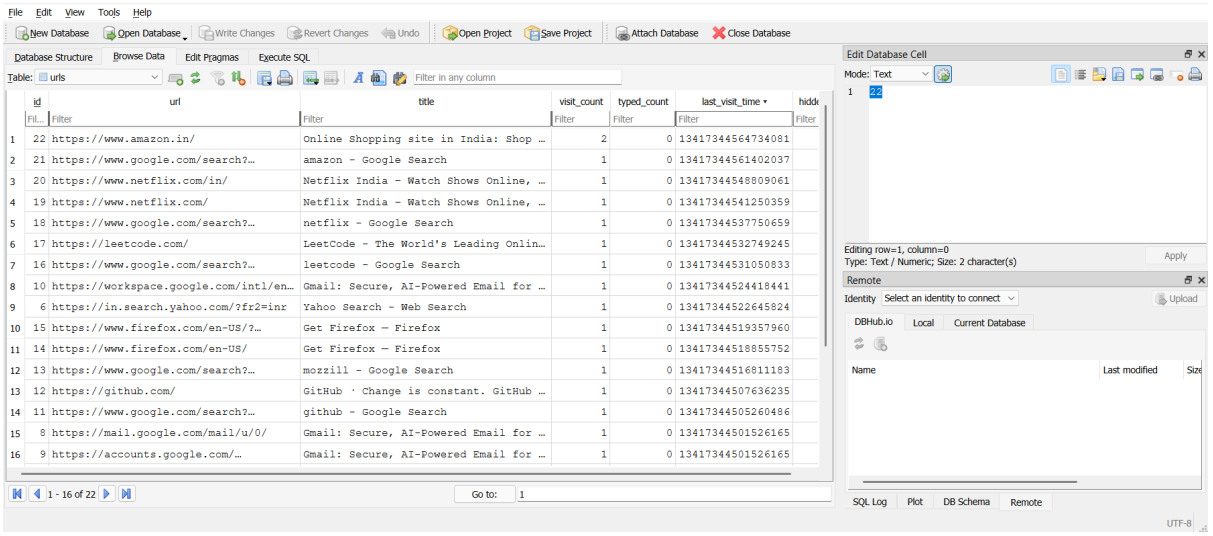
Question10: Which website was visited the most number of times? Observe the column named visit_count.



	id	url	title	visit_count
1	22	https://www.amazon.in/	Online Shopping site in India: Shop …	2
2	1	https://www.google.com/search?…	youtube - Google Search	1
3	2	https://www.google.com/search?…	youtube - Google Search	1
4	3	https://www.youtube.com/	YouTube	1
5	4	https://www.google.com/search?…	yahoo - Google Search	1
6	5	https://in.yahoo.com/	Yahoo Search - Web Search	1
7	6	https://in.search.yahoo.com/?fr2=inr	Yahoo Search - Web Search	1
8	7	https://www.google.com/search?…	gmail - Google Search	1
9	8	https://mail.google.com/mail/u/0/	Gmail: Secure, AI-Powered Email for …	1
10	9	https://accounts.google.com/…	Gmail: Secure, AI-Powered Email for …	1
11	10	https://workspace.google.com/intl/en…	Gmail: Secure, AI-Powered Email for …	1
12	11	https://www.google.com/search?…	github - Google Search	1
13	12	https://github.com/	GitHub · Change is constant. GitHub …	1
14	13	https://www.google.com/search?…	mozzill - Google Search	1
15	14	https://www.firefox.com/en-US/	Get Firefox — Firefox	1
16	15	https://www.firefox.com/en-US/?…	Get Firefox — Firefox	1

Answer: The website with the highest number of visits is https://www.amazon.in/, recorded with a total visit count of 2.

Question11: Which website was visited most recently? Review the last_visit_time column.

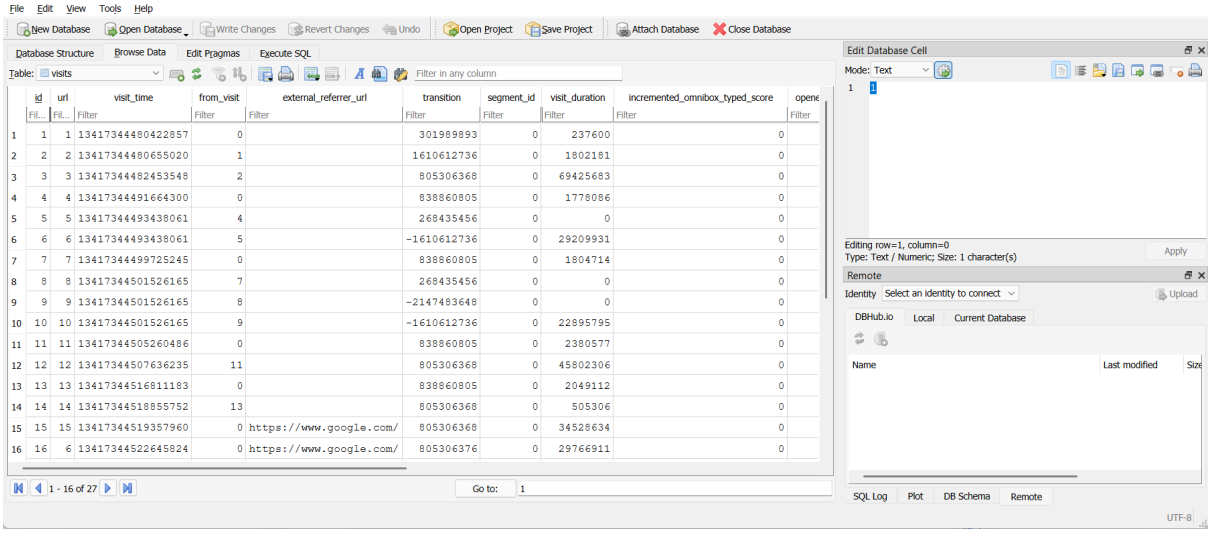


	id	url	title	visit_count	typed_count	last_visit_time	hidde
1	22	https://www.amazon.in/	Online Shopping site in India: Shop …	2	0	13417344564734081	
2	21	https://www.google.com/search?…	amazon - Google Search	1	0	13417344561402037	
3	20	https://www.netflix.com/in/	Netflix India - Watch Shows Online, …	1	0	13417344548809061	
4	19	https://www.netflix.com/	Netflix India - Watch Shows Online, …	1	0	13417344541250359	
5	18	https://www.google.com/search?…	netflix - Google Search	1	0	13417344537750659	
6	17	https://leetcode.com/	LeetCode - The World's Leading Onlin…	1	0	13417344532749245	
7	16	https://www.google.com/search?…	leetcode - Google Search	1	0	13417344531050833	
8	10	https://workspace.google.com/intl/en…	Gmail: Secure, AI-Powered Email for …	1	0	13417344524418441	
9	6	https://in.search.yahoo.com/?fr2=inr	Yahoo Search - Web Search	1	0	13417344522645824	
10	15	https://www.firefox.com/en-US/?…	Get Firefox — Firefox	1	0	13417344519357960	
11	14	https://www.firefox.com/en-US/	Get Firefox — Firefox	1	0	13417344518855752	
12	13	https://www.google.com/search?…	mozzill - Google Search	1	0	13417344516811183	
13	12	https://github.com/	GitHub · Change is constant. GitHub …	1	0	13417344507636235	
14	11	https://www.google.com/search?…	github - Google Search	1	0	13417344505260486	
15	8	https://mail.google.com/mail/u/0/	Gmail: Secure, AI-Powered Email for …	1	0	13417344501526165	
16	9	https://accounts.google.com/…	Gmail: Secure, AI-Powered Email for …	1	0	13417344501526165	

Answer: https://www.amazon.in/ is the most recently visited website.

31)Now select the table named visits.
32)Examine the records in the visits table.

Question12: How many browsing records are present in the visits table?

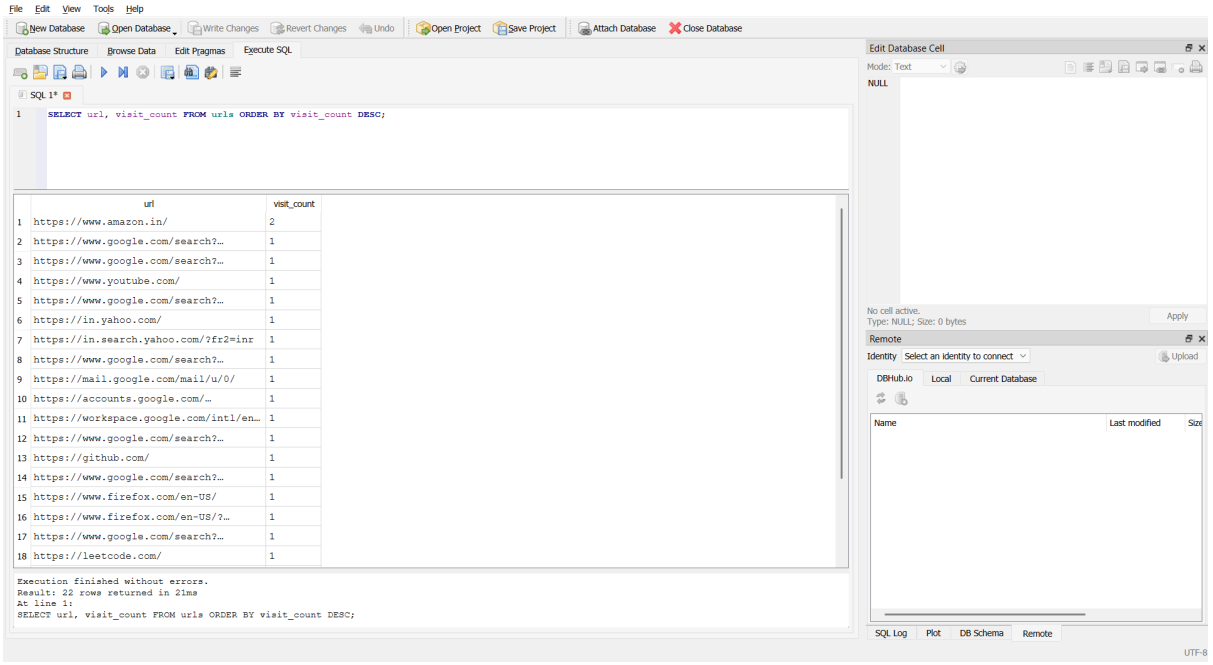


	id	url	visit_time	from_visit	external_referrer_url	transition	segment_id	visit_duration	incremented_omnibox_typed_score	opene
1	1	1	13417344480422857	0		301989893	0	237600	0	
2	2	2	13417344480655020	1		1610612736	0	1802181	0	
3	3	3	13417344482453548	2		805306368	0	69425683	0	
4	4	4	13417344491664300	0		838860805	0	1778086	0	
5	5	5	13417344493438061	4		268435456	0	0	0	
6	6	6	13417344493438061	5		-1610612736	0	29209931	0	
7	7	7	13417344499725245	0		838860805	0	1804714	0	
8	8	8	13417344501526165	7		268435456	0	0	0	
9	9	9	13417344501526165	8		-2147483648	0	0	0	
10	10	10	13417344501526165	9		-1610612736	0	22895795	0	
11	11	11	13417344505260486	0		838860805	0	2380577	0	
12	12	12	13417344507636235	11		805306368	0	45802306	0	
13	13	13	13417344516811183	0		838860805	0	2049112	0	
14	14	14	13417344518855752	13		805306368	0	505306	0	
15	15	15	13417344519357960	0	https://www.google.com/	805306368	0	34528634	0	
16	16	6	13417344522645824	0	https://www.google.com/	805306376	0	29766911	0	

Answer: There are in total 27 browsing records present in the visits table.

33)Click on the Execute SQL tab.
34)Run the following example query:

SELECT url, visit_count FROM urls ORDER BY visit_count DESC;



Question13: Which are the top 4 most visited websites according to the database? Review the results returned by the query.

Answer: The top 4 most visited websites according to the database are:

- https://www.amazon.in/
- https://www.google.com/search?q=youtube&oq=yout&gs_lcrp=EgZjaHJvbWUqEAgAEAAYgwEY4wIYsQMYgAQyEAgAEAAYgwEY4wIYsQMYgAQyDQgBEC4YgwEYsQMYgAQyBggCEEUYOTINCAMQABiDARixAxiABDIKCAQQABixAxiABDIKCAUQABixAxiABDIKCAYQLhixAxiABDIGCAcQBRhA0gEHNjkzajBqN6gCCLACAfEFsF3cdMKczE7xBbBd3HTCnMxO&sourceid=chrome&ie=UTF-8
- https://www.google.com/search?q=youtube&oq=yout&gs_lcrp=EgZjaHJvbWUqEAgAEAAYgwEY4wIYsQMYgAQyEAgAEAAYgwEY4wIYsQMYgAQyDQgBEC4YgwEYsQMYgAQyBggCEEUYOTINCAMQABiDARixAxiABDIKCAQQABixAxiABDIKCAUQABixAxiABDIKCAYQLhixAxiABDIGCAcQBRhA0gEHNjkzajBqN6gCCLACAfEFsF3cdMKczE7xBbBd3HTCnMxO&sourceid=chrome&ie=UTF-8&sei=4NyradijFsqwwcsP_8iluQw
- https://www.youtube.com/